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(54) **SYSTEMS AND METHODS FOR KNEE JOINT THERAPY**

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(51) **Int. Cl.**
A61H 1/02 (2006.01)

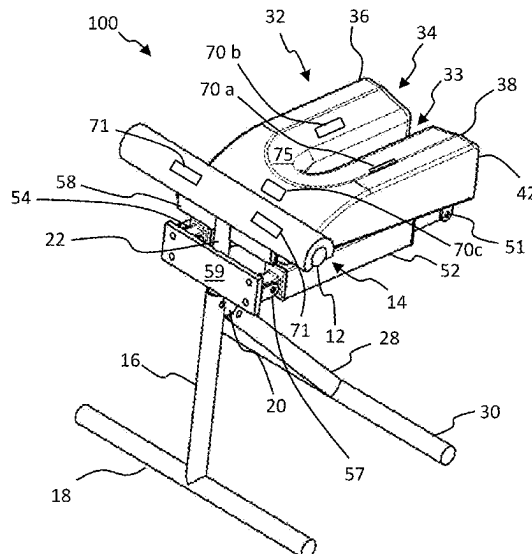
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CPC **A61H 1/024** (2013.01); **A61H 2201/0119**
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CPC A61H 1/024; A61H 2201/0119; A61H
2201/0157; A61H 2201/0192; A61H
2201/1238; A61H 2201/1292; A61H
2201/1409; A61H 2201/164; A61H
2201/1676; A61H 2201/169; A61H
2201/5007; A61H 2201/5025; A61H

(57) **ABSTRACT**

A knee joint rehabilitation apparatus may include a support bar fixed to a height-adjustment bar, the support bar being configured to receive a knee joint posterior; a plurality of pads configured to removably mount to the support bar to adjust an effective outer diameter of the support bar; and a lower leg support connected to the height-adjustment bar, the lower leg support being mounted to the height-adjustment bar so as to allow for changes in an angular position between said lower leg support and said height-adjustment bar.

20 Claims, 20 Drawing Sheets



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See application file for complete search history.

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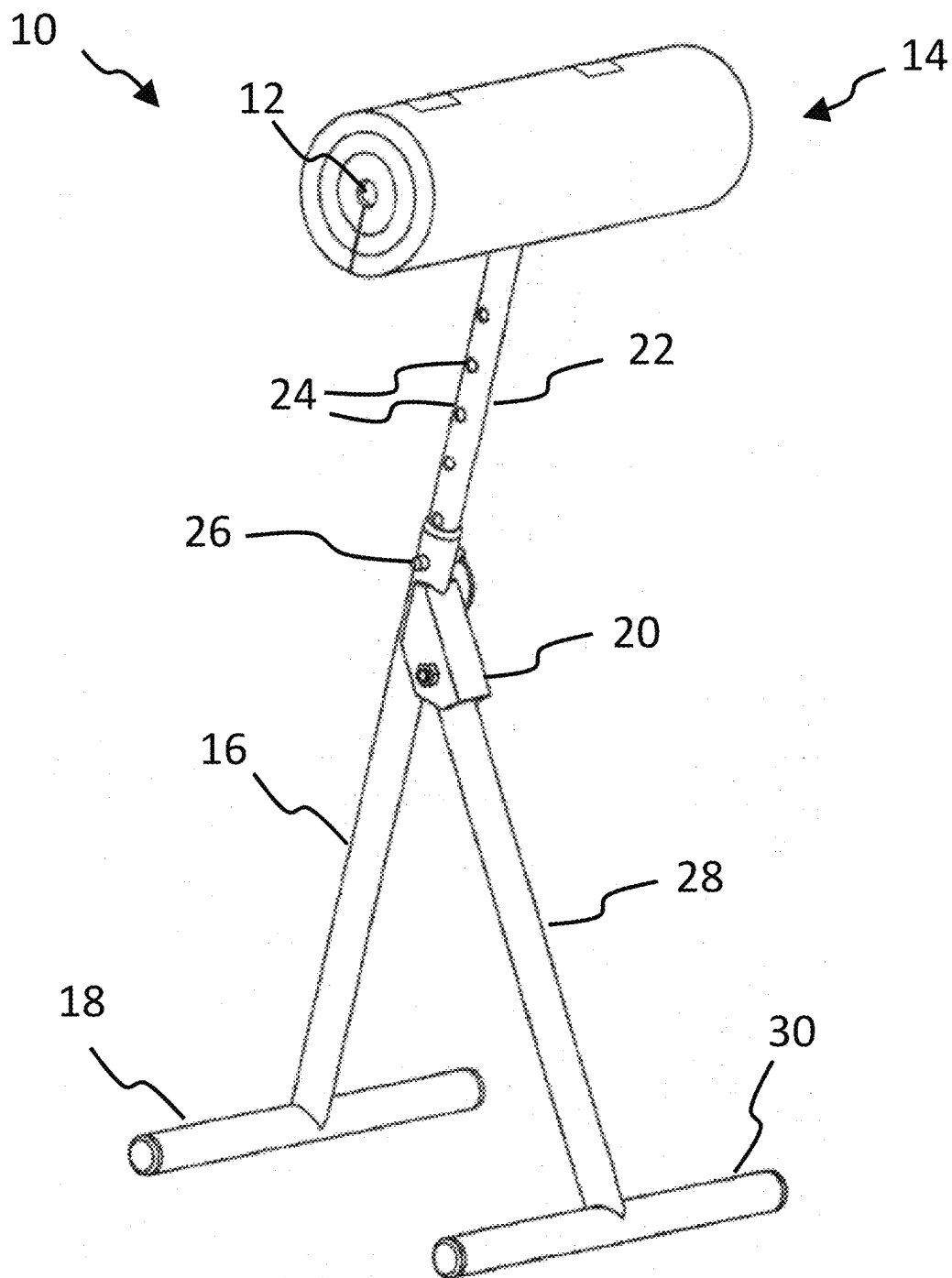


Fig. 1

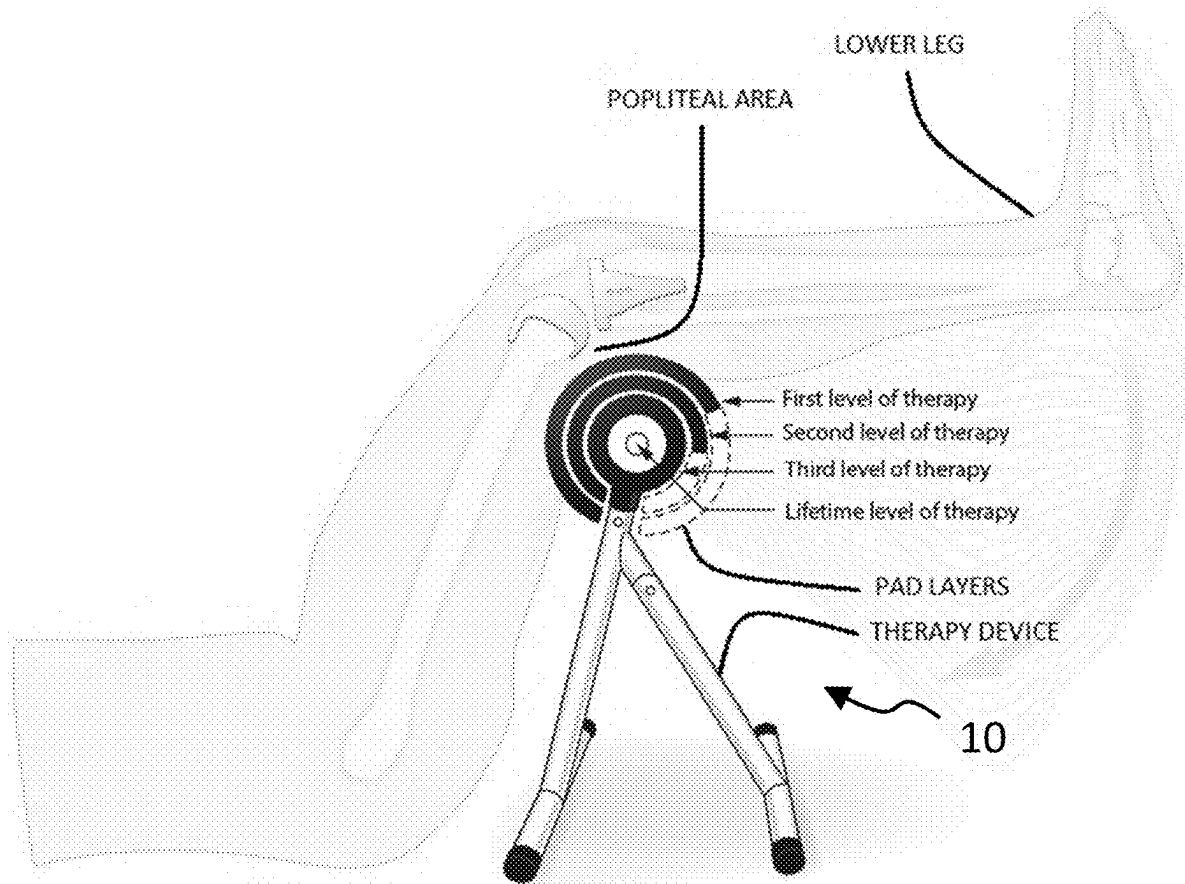


Fig. 2

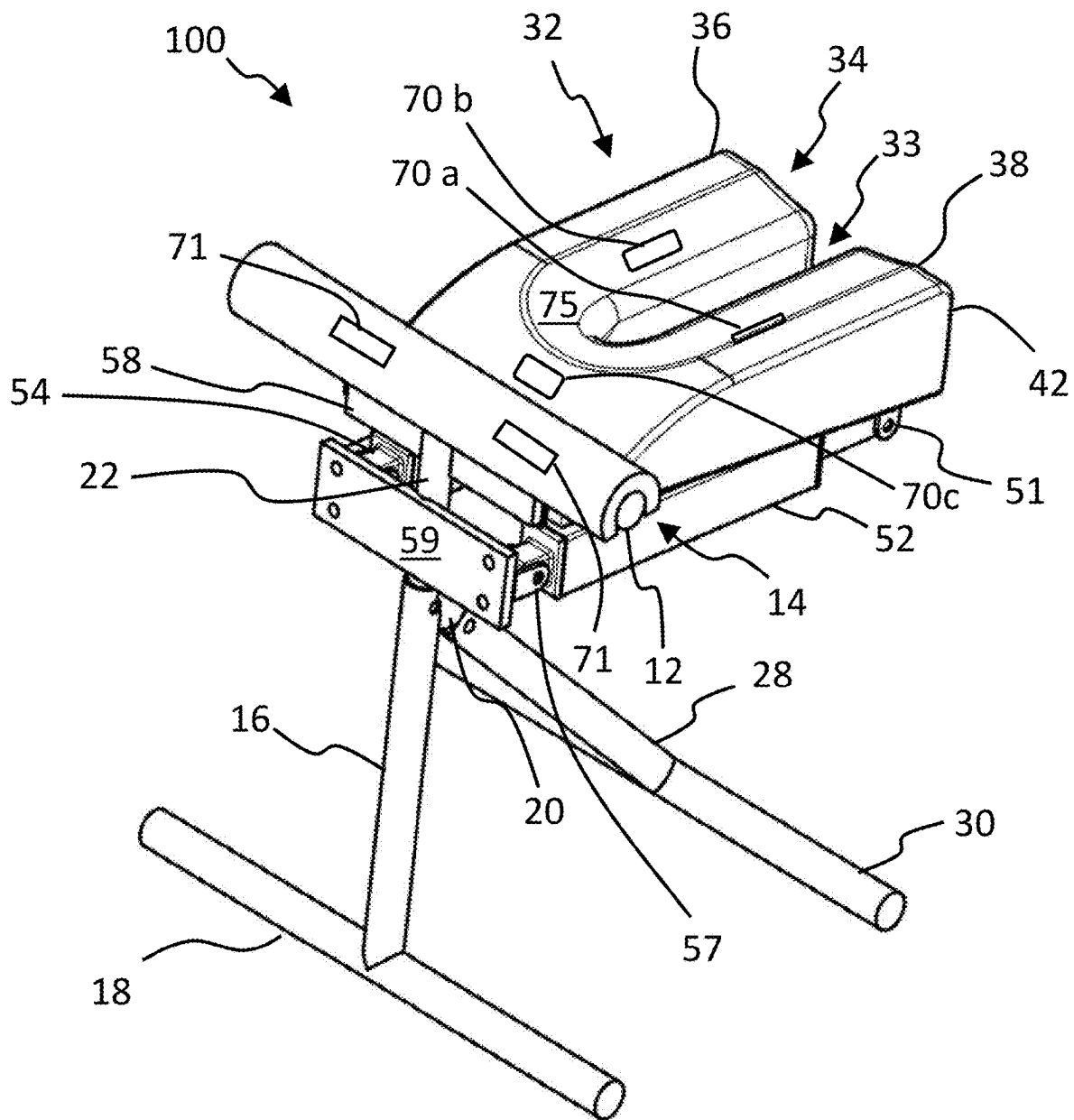


Fig. 3

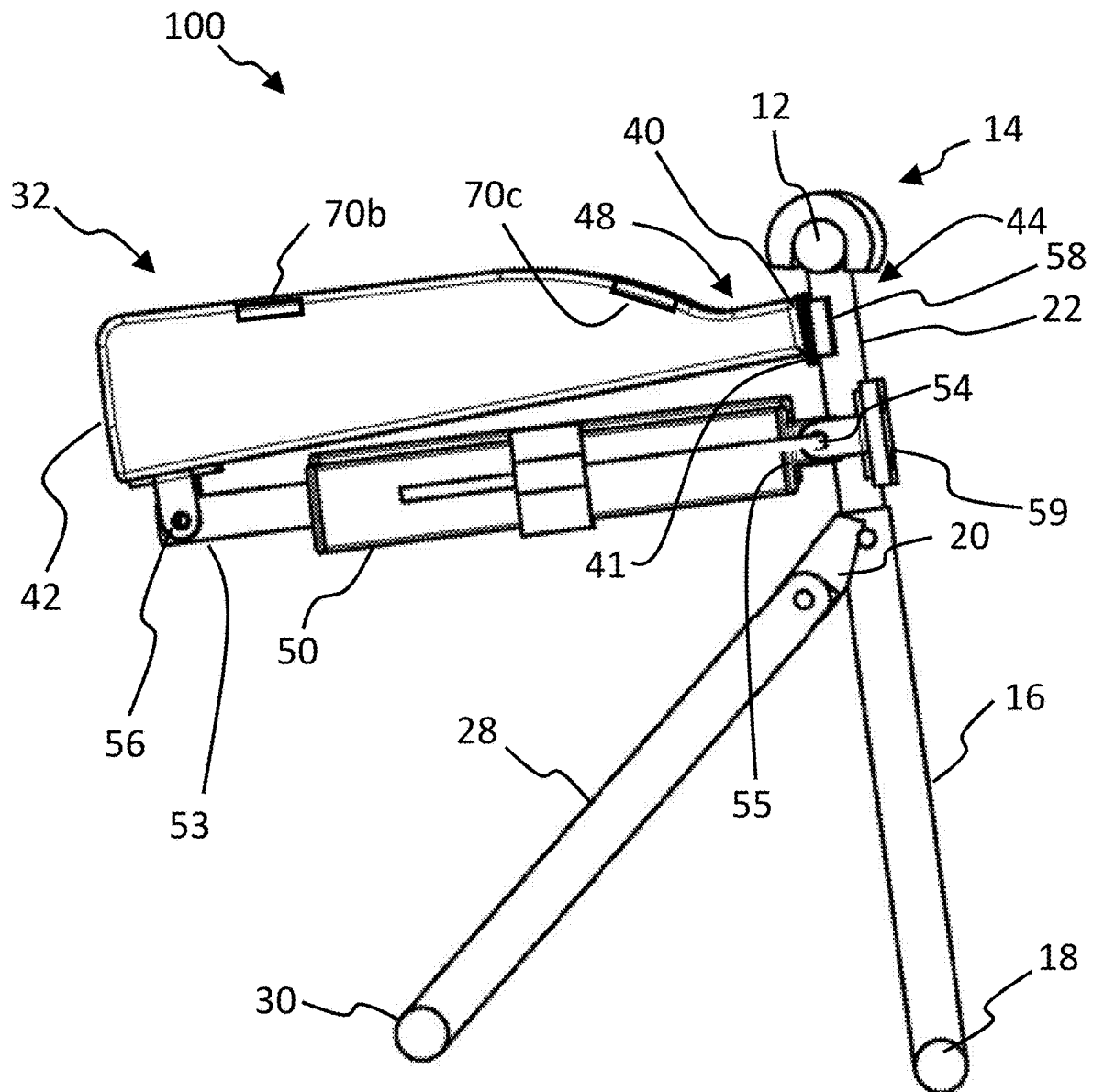


Fig. 4

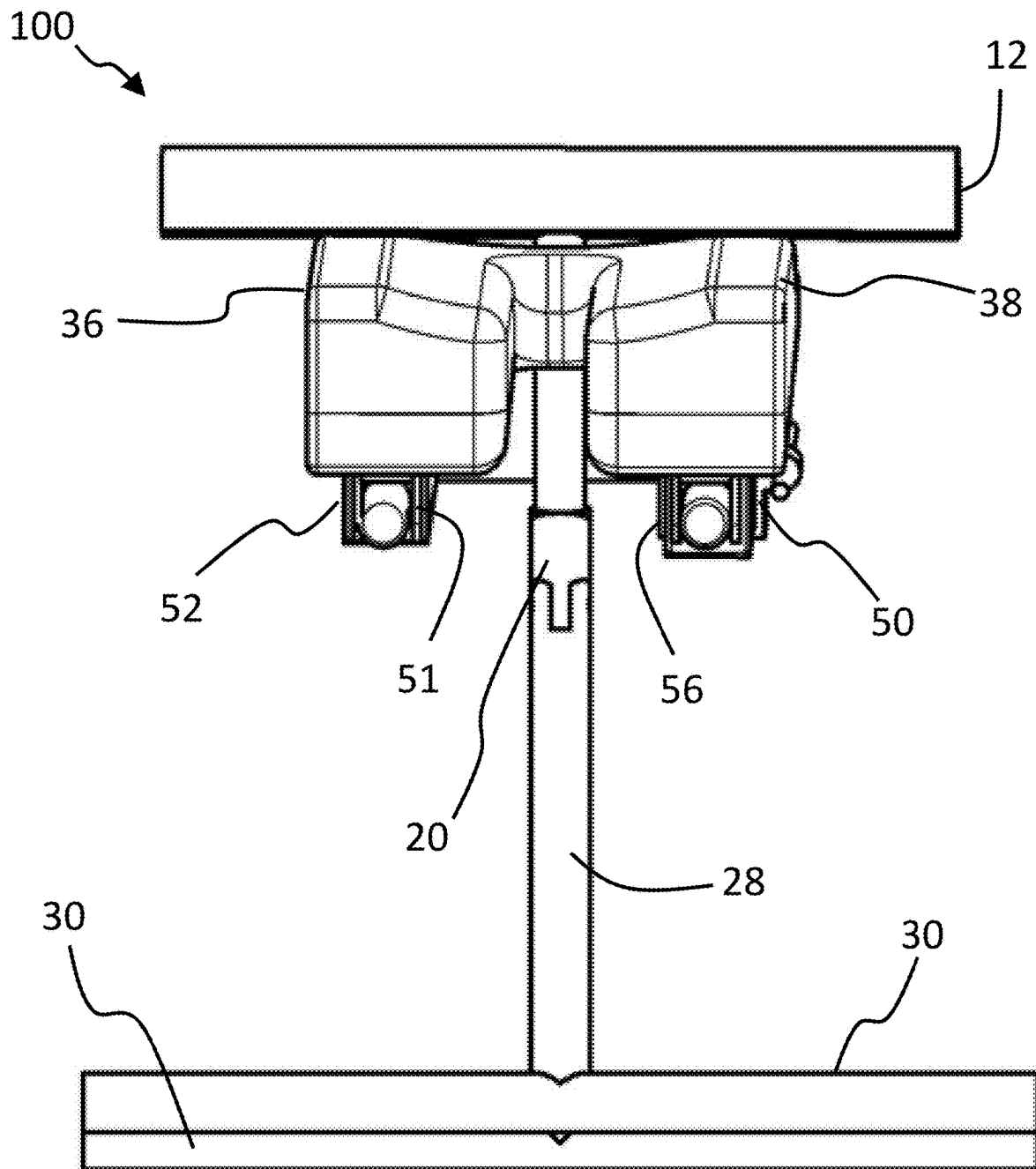


Fig. 5

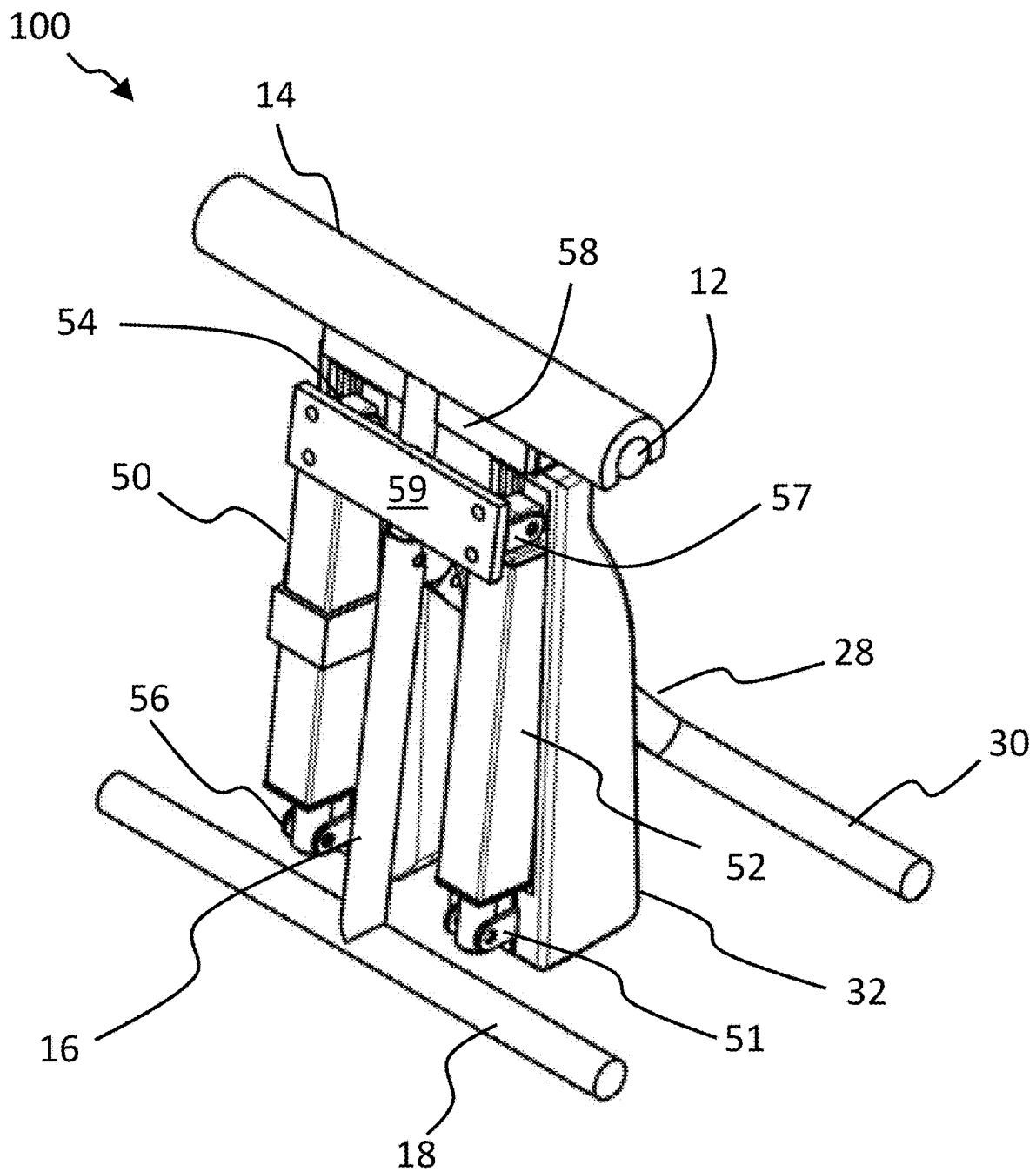


Fig. 6

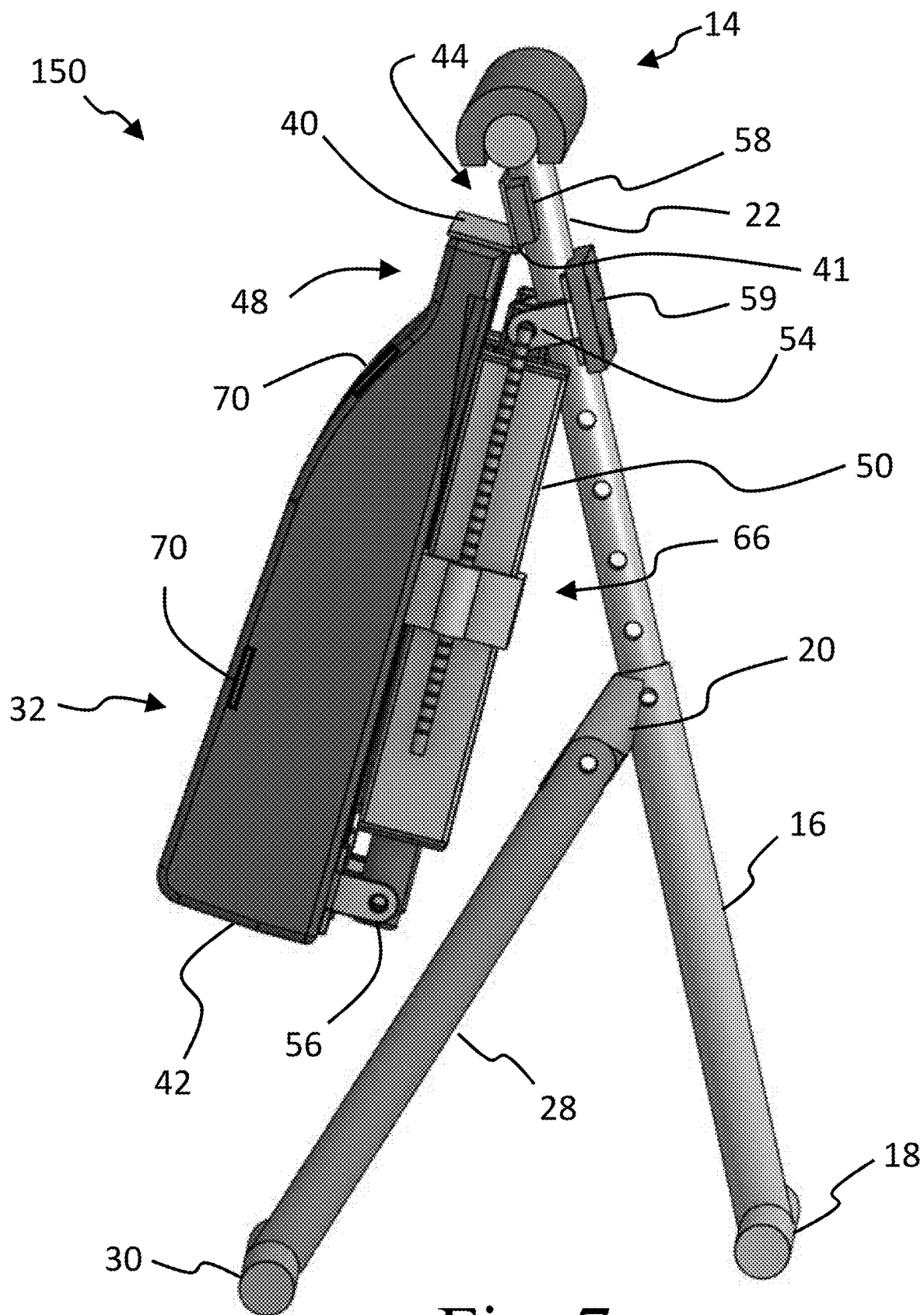


Fig. 7

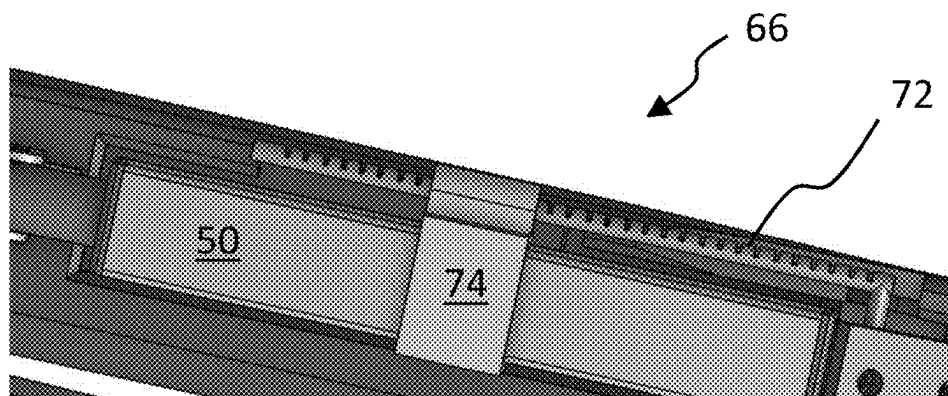


Fig. 8

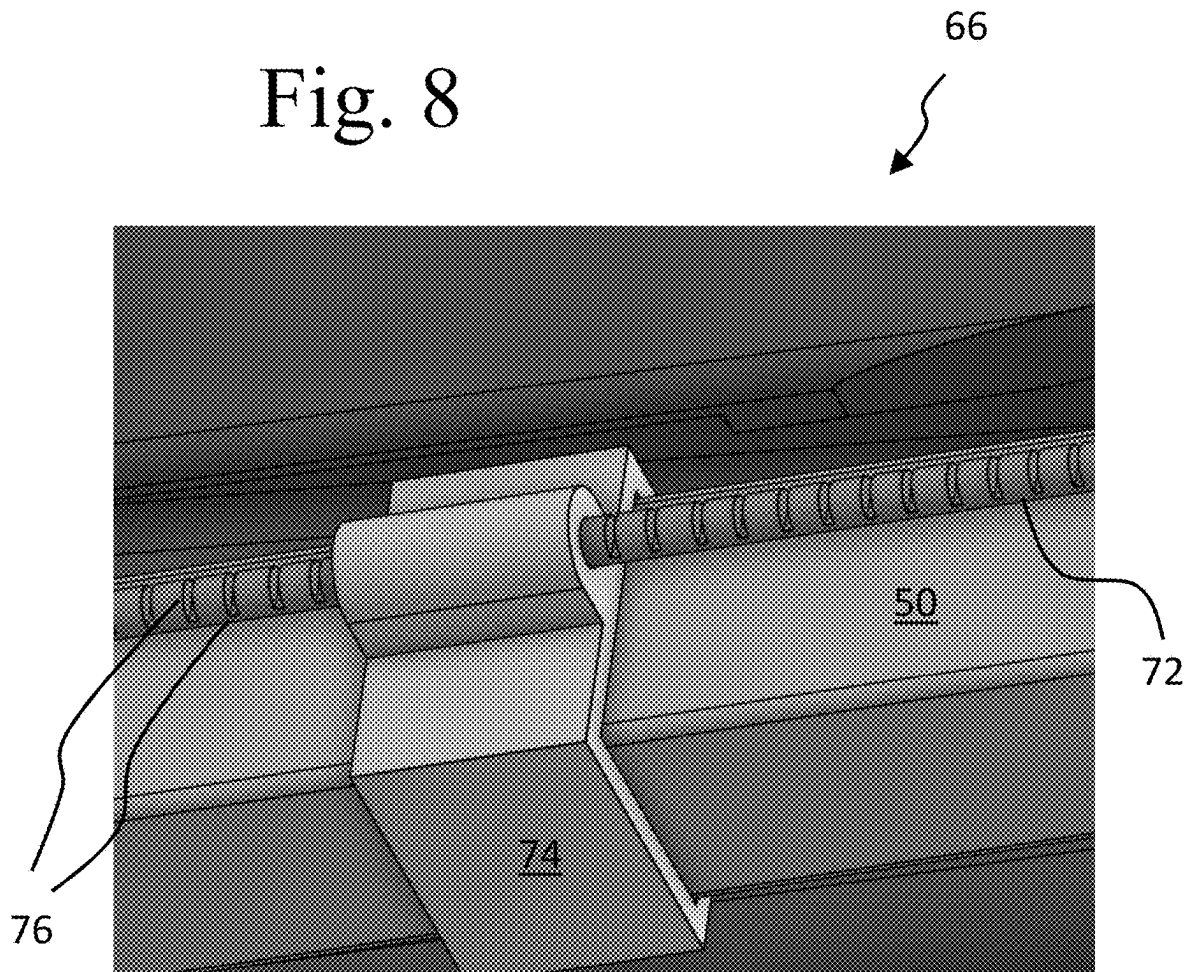


Fig. 9

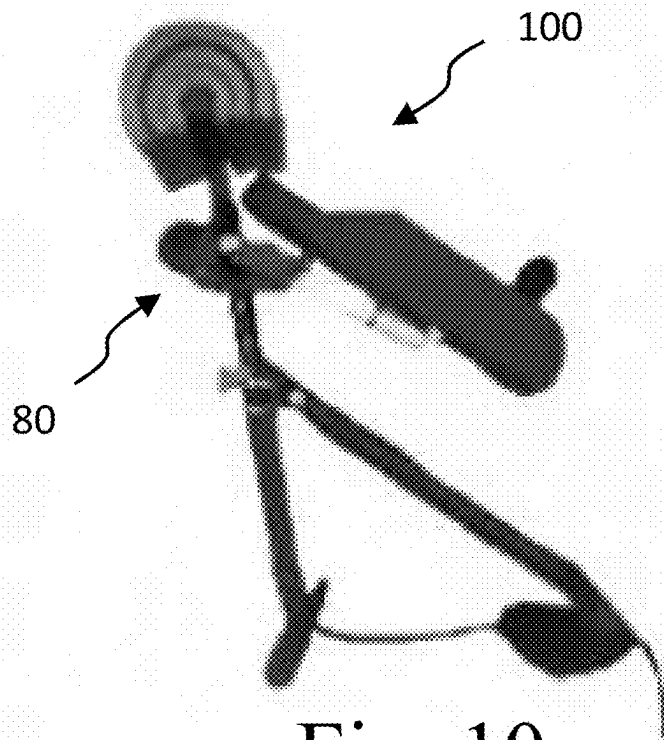


Fig. 10

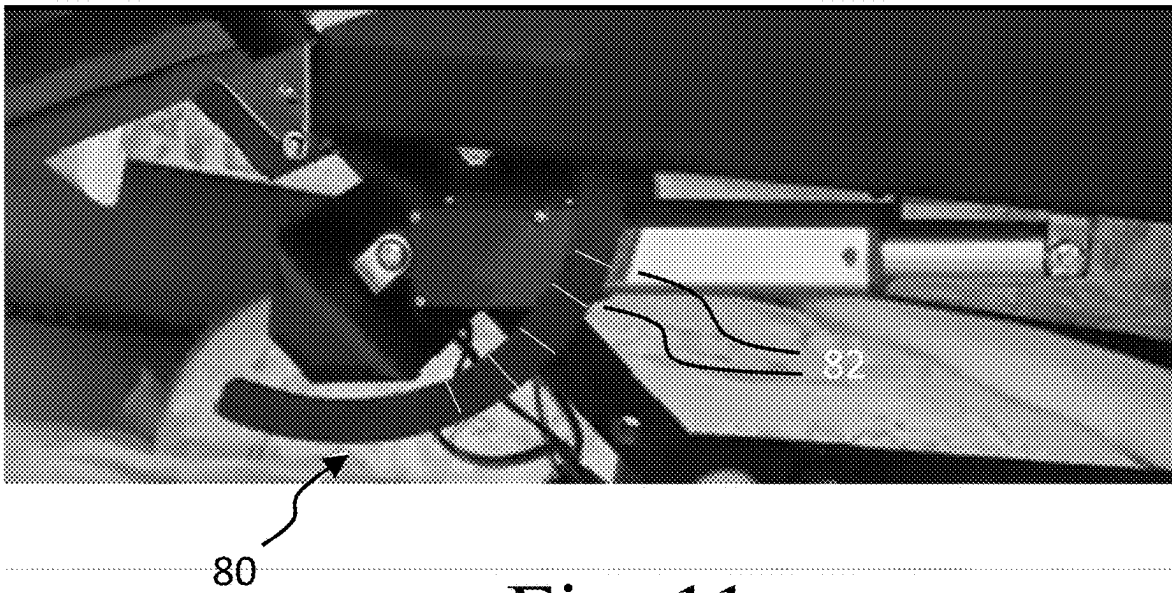


Fig. 11

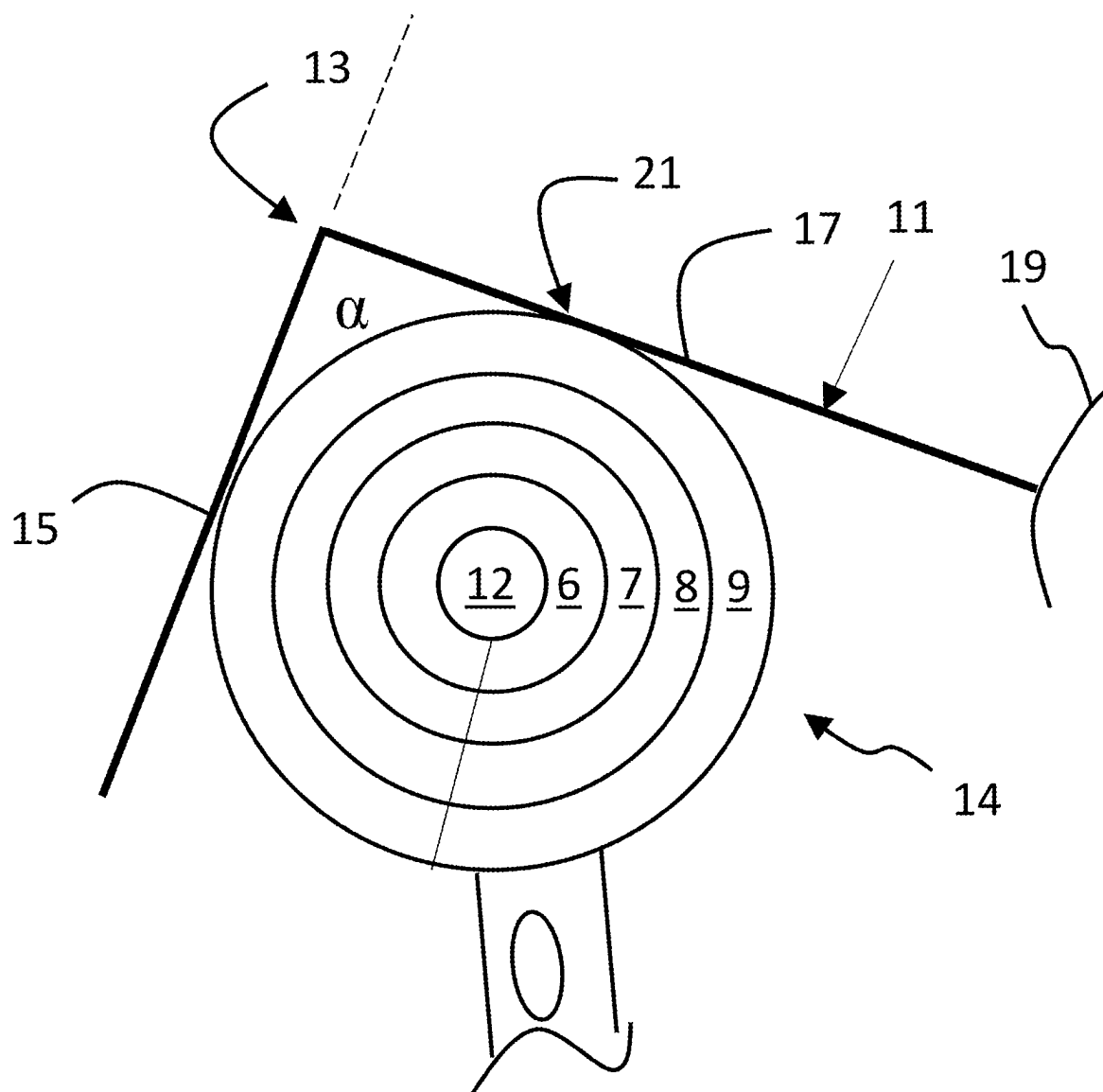


Fig. 12

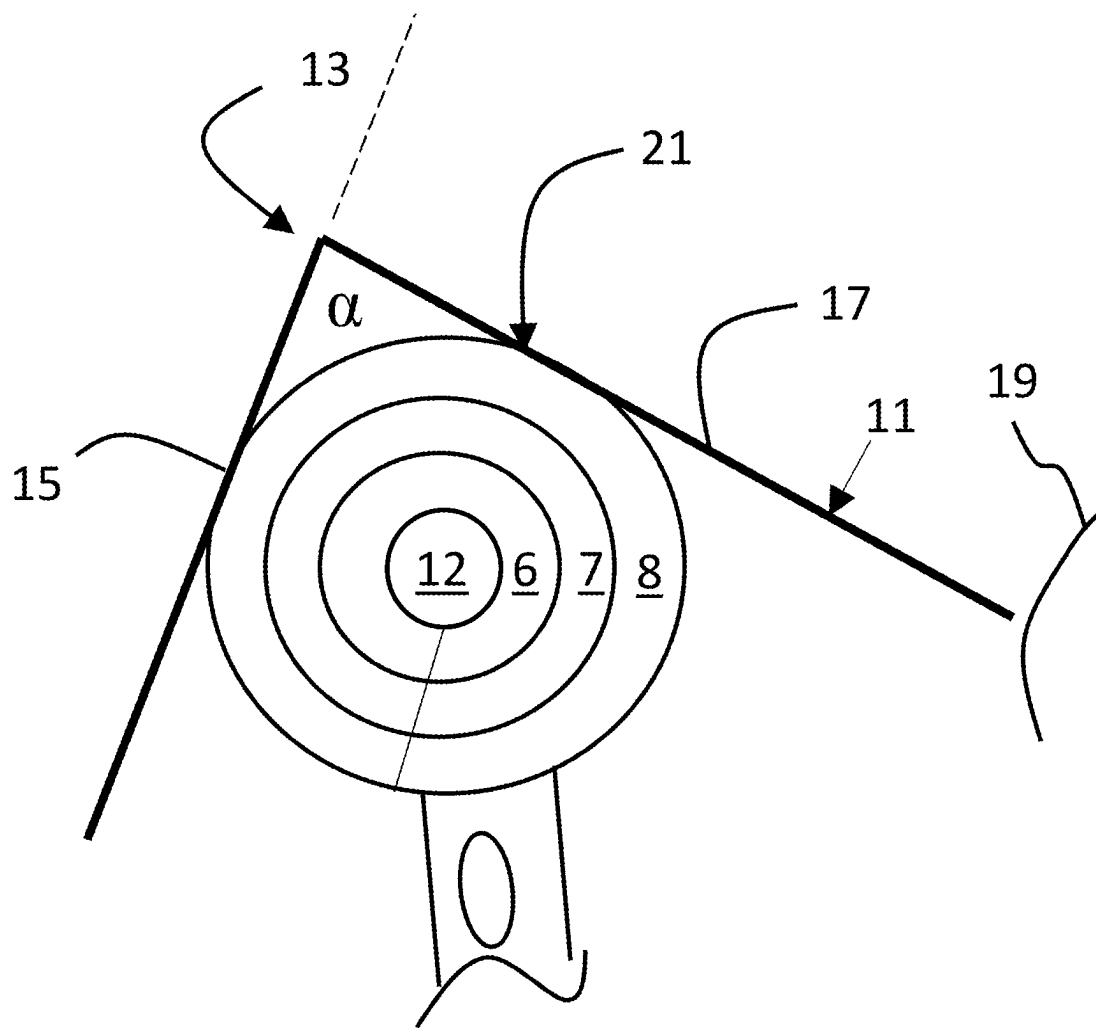


Fig. 13

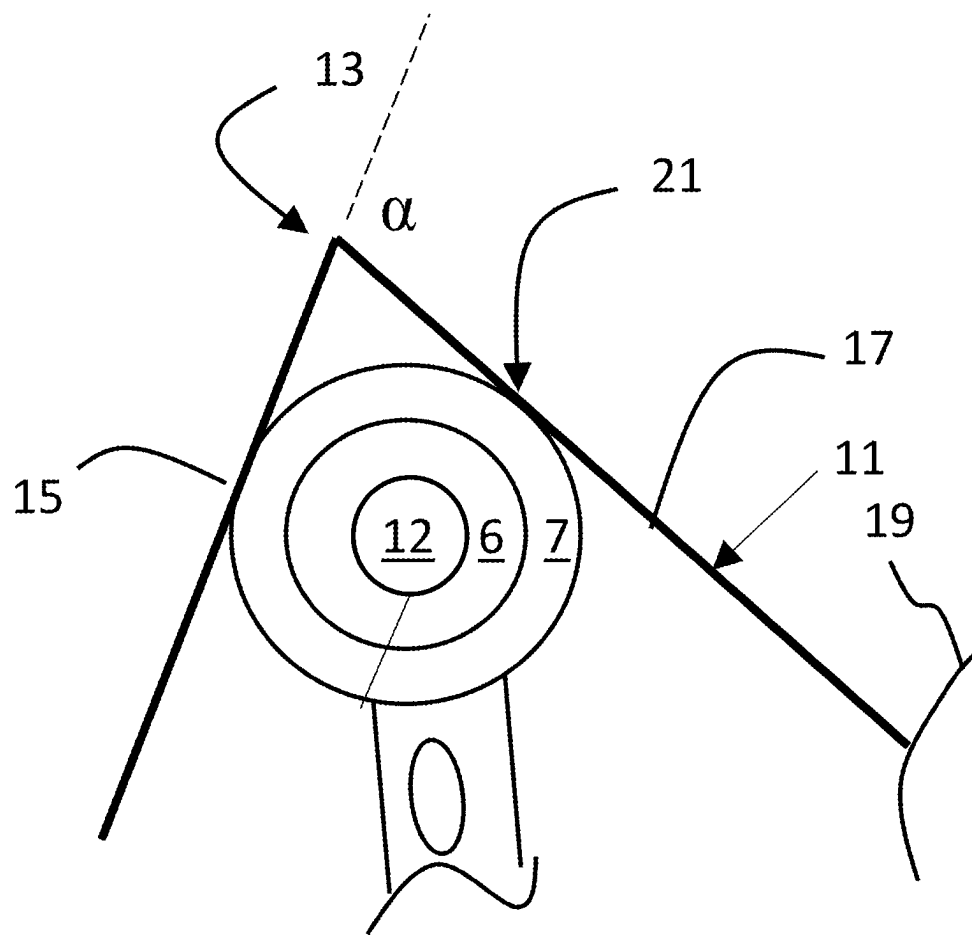


Fig. 14

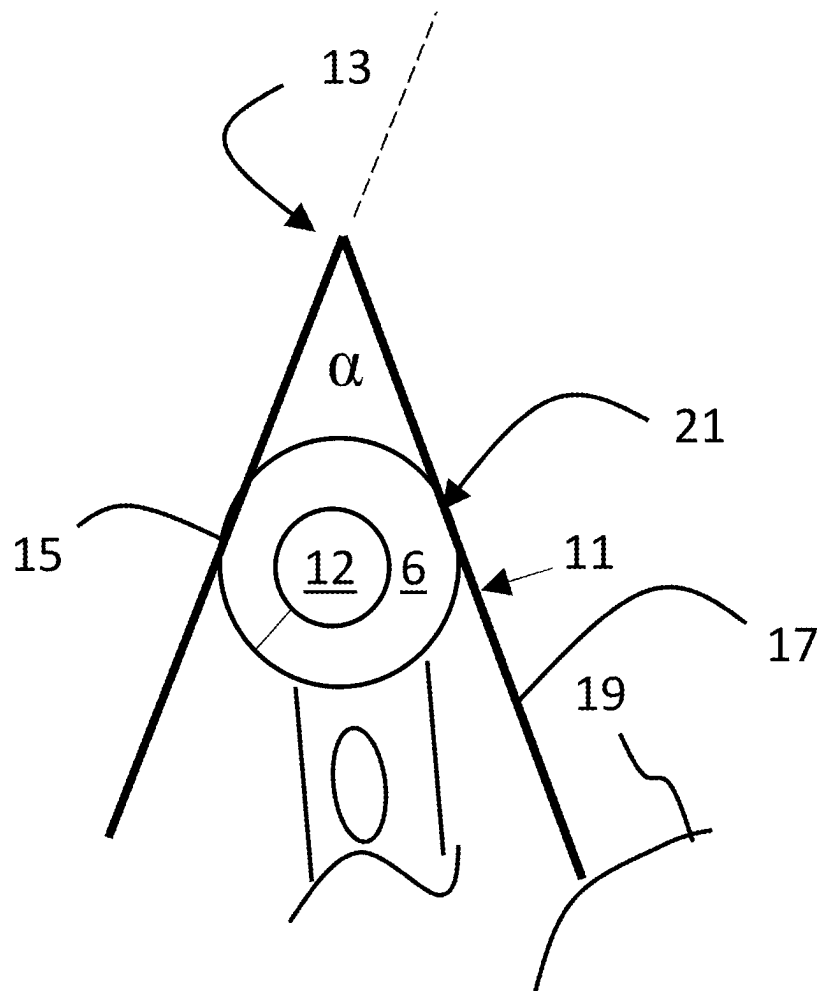


Fig. 15

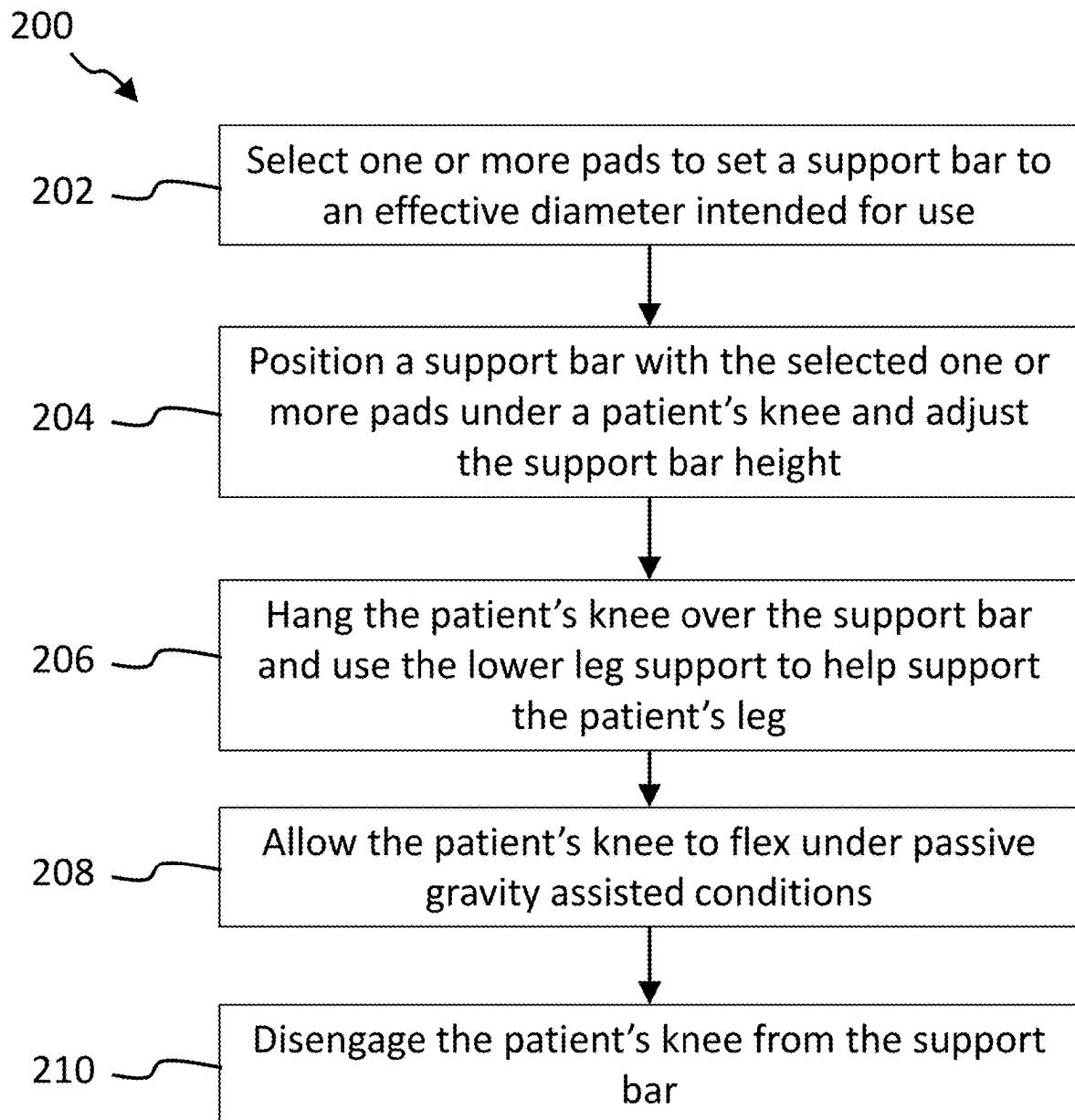


Fig. 16

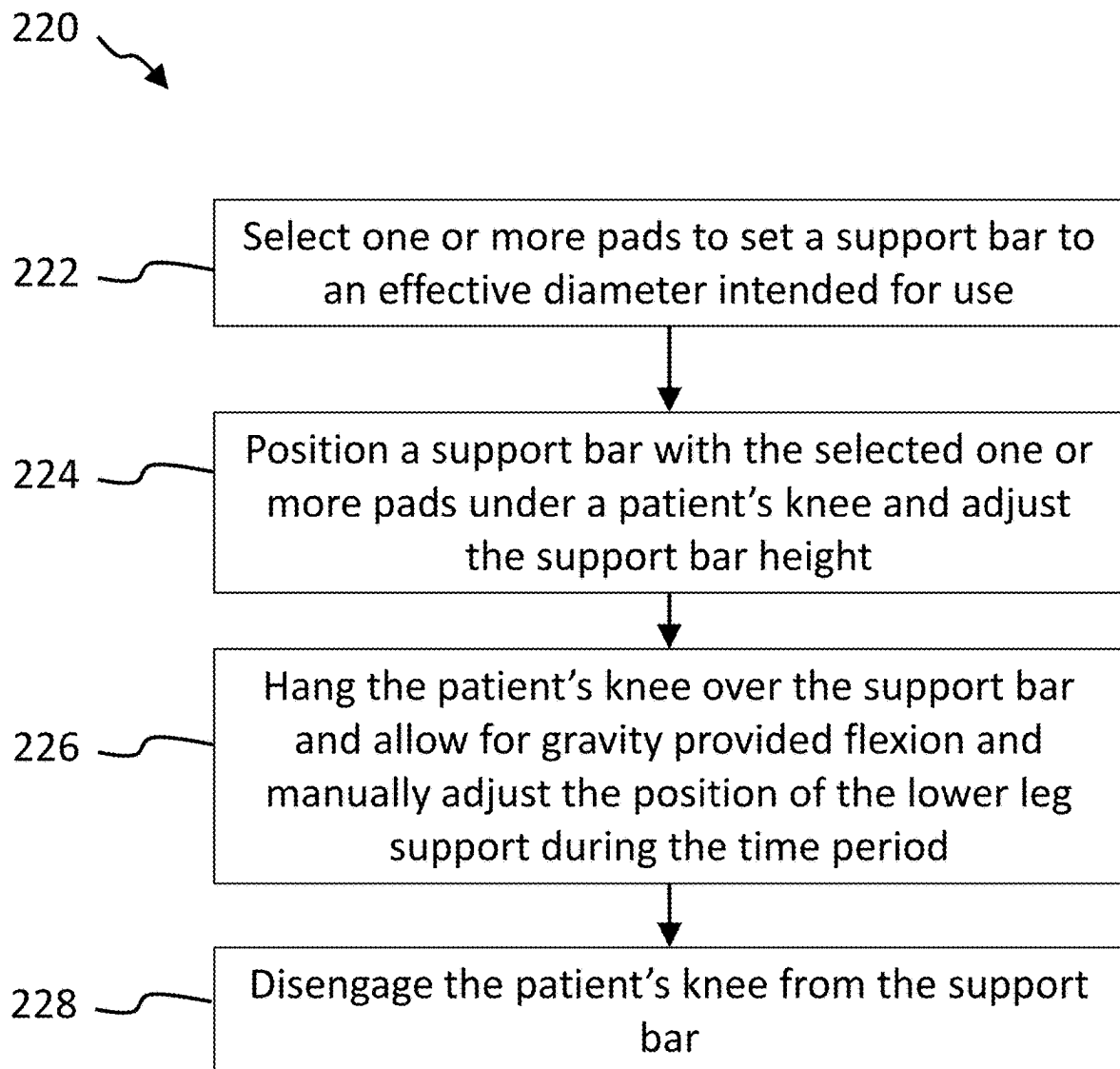


Fig. 17

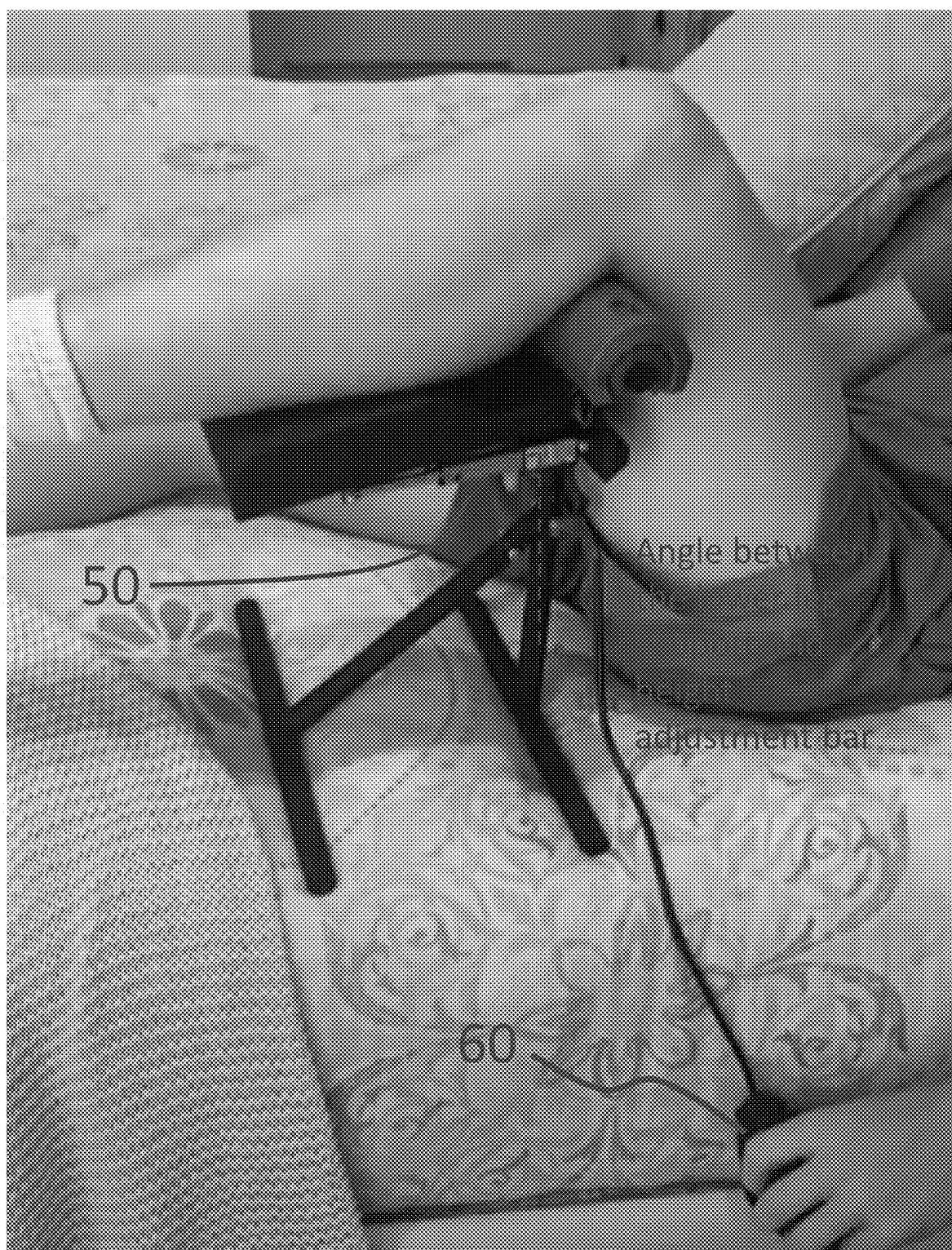


Fig. 18

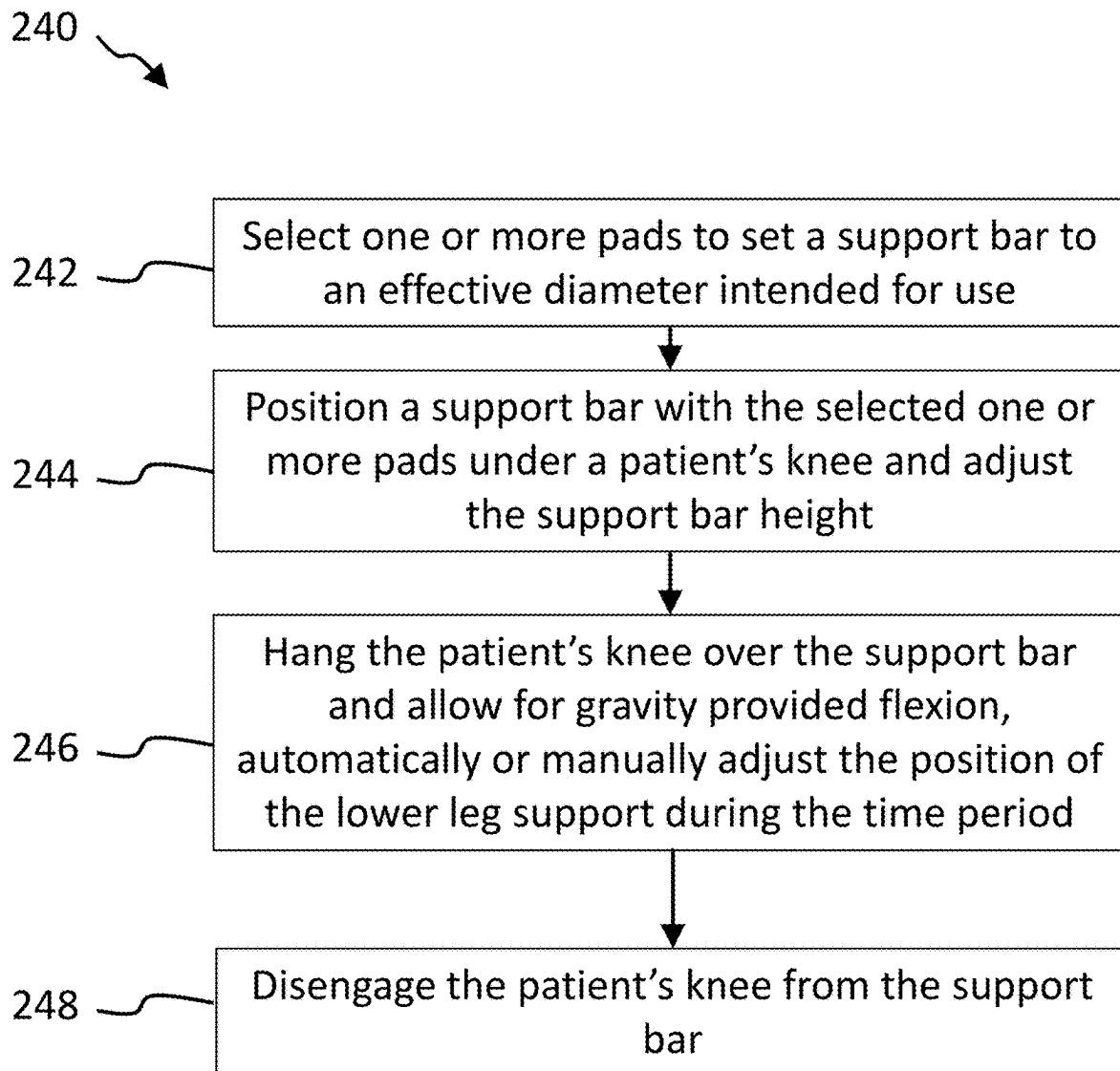


Fig. 19



Fig. 20

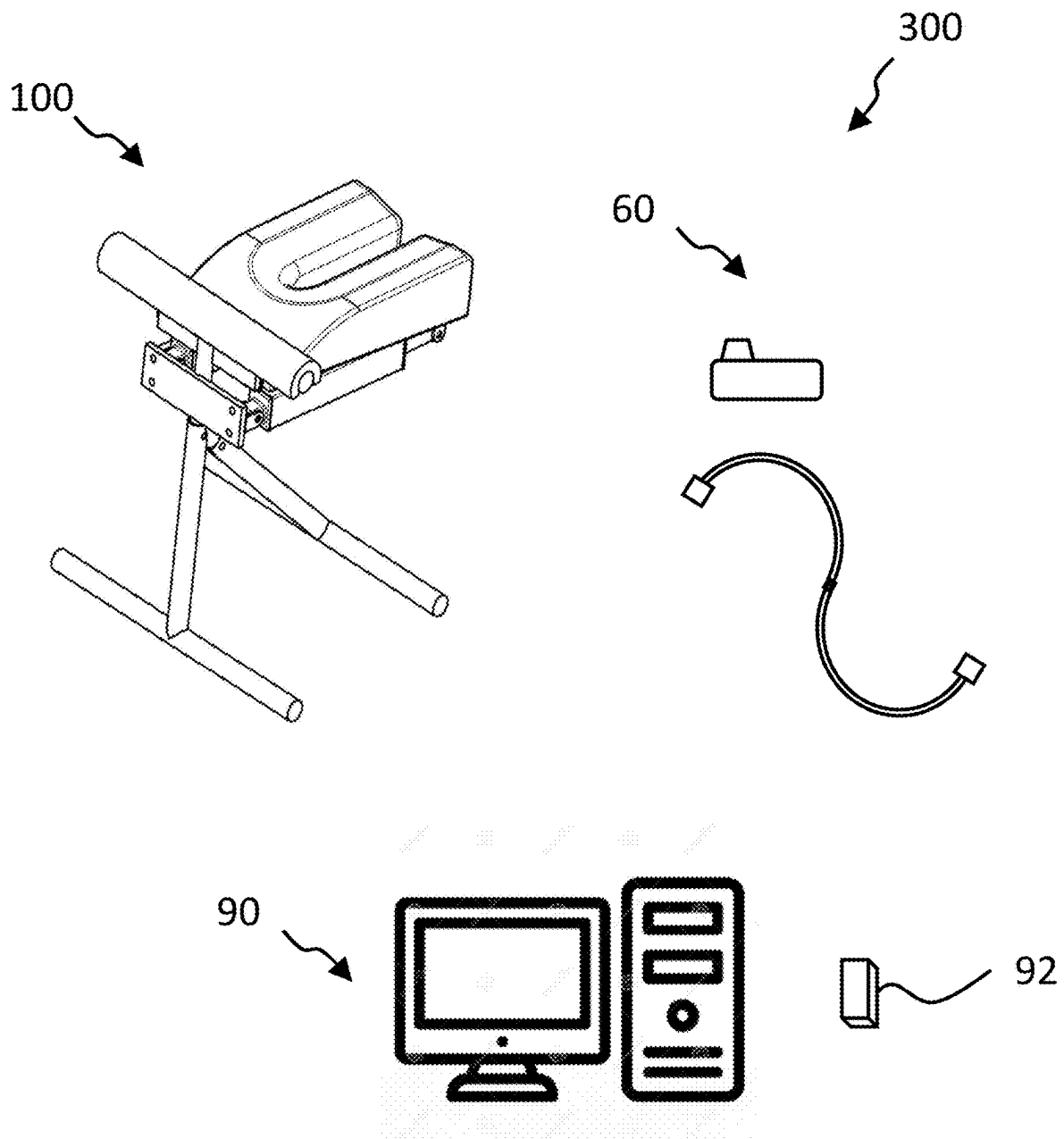


Fig. 21

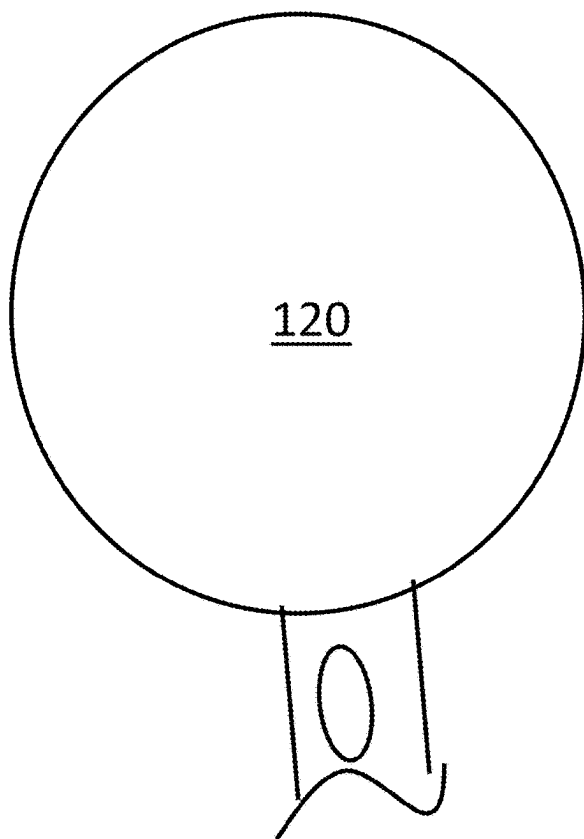


Fig. 22

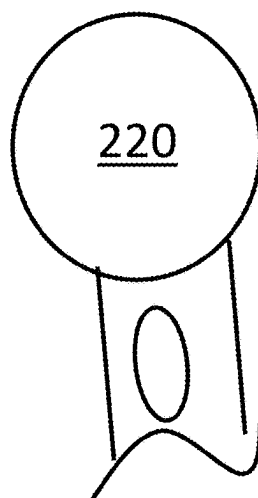


Fig. 23

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SYSTEMS AND METHODS FOR KNEE JOINT THERAPY

CROSS REFERENCE TO RELATED APPLICATIONS

This application claims priority to U.S. Provisional Patent Application No. 63/399,938 filed Aug. 22, 2022, titled “Systems and Methods for Knee Joint Therapy”, the disclosure of which is herein fully incorporated by reference.

FIELD

This application relates generally to physical therapy systems and methods for treatment of a knee joint.

BACKGROUND

The anatomy of a knee joint includes three bones, namely, the femur (thighbone), the tibia (shinbone) and the patella (kneecap). The patella is located in front of the knee joint and provides partial protection for the anterior portion of the knee. The knee joint includes a number of ligaments. Collateral ligaments are located vertically on opposed parallel sides of the knee joint. The medial collateral ligament is located on the outer inside of the knee joint and links the femur and tibia. The lateral collateral ligament is located on the outer outside of the knee joint, and also links the femur and tibia. The purpose of the collateral ligaments is towards controlling knee joint sideways motion, thus fortifying the knee joint against unusual movement.

Cruciate ligaments are located inside of the knee joint and cross each other in the general form of an “X”. The anterior cruciate ligament is located in front, behind the patella, and the posterior cruciate ligament is located in the back, linking the femur and tibia.

Upon completion of knee joint surgery or other type of knee joint treatment, such as knee joint replacement or replacement or repair of any torn or damaged previously described ligaments, bone fracture and bone fracture surgery, or internal fixation surgery, or for treatment of arthritic conditions, or meniscus repair, or for knee injury recovery, regaining normal knee joint flexibility, range of motion and strength typically requires a physical therapy rehabilitation program.

Following knee surgery, or arthritic treatment, or injury treatment, or other procedure or course of treatment for the knee joint, physical therapy may be beneficially applied toward restoring full and painless motion of the knee joint, along with re-establishing associated ligament and muscle suppleness. Restoration may be followed by a knee joint strengthening program focused on regaining normal function of the knee joint. A physical therapy program progressively stretches muscle(s) and tendon(s) to their former flexibility. A final rehabilitation phase encourages complete return of normal knee joint functionality.

There remains a need for a system and method for a knee joint flexion and extension support allowing for incrementally-applied physical therapy.

SUMMARY

In some embodiments, a knee joint rehabilitation apparatus may include a support bar fixed to a height-adjustment bar, the support bar being configured to receive a knee joint posterior; a plurality of pads configured to removably mount to the support bar to adjust an effective outer diameter of the

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support bar; and a lower leg support connected to the height-adjustment bar, the lower leg support being mounted to the height-adjustment bar so as to allow for changes in an angular position between said lower leg support and said height-adjustment bar

It is an objective of some embodiments herein to provide a knee joint rehabilitation device configured for knee joint flexion and to allow a patient’s knee joint to progress through different levels of flexion using gravity assisted therapeutic methods. To progress through different stages or levels of therapy, a support bar and system of pads may be used with the patient’s leg extended over the support bar. With the leg disposed on the support bar, the physical therapist may allow gravity to pull the lower leg downwards through a range of motion to help stretch the knee joint tissues. The lower leg may effectively serve as a lever, with the femur and patient’s body weight at a knee side of a fulcrum point serving as an anchor weight. The gravity force serves to gently distract the knee joint, thus allowing the knee joint to separate and stretch substantially without creating compressive forces in the joint. In some embodiments, at least a portion of the weight of the patient’s leg may be supported by a lower leg support. The lower leg support may, for example, be slowly adjusted in position so that a decreasing proportion of the weight of the patient’s leg is supported thereby. This may, for example, allow a patient to transition between incremental levels of knee joint flexion more gently.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a perspective view of an embodiment of a knee joint rehabilitation device or apparatus.

FIG. 2 is a schematic view showing a user’s knee positioned on the knee joint rehabilitation device of FIG. 1.

FIG. 3 is a perspective view of another embodiment of a knee joint rehabilitation device which includes a lower leg support.

FIG. 4 is a side view of the knee joint rehabilitation device of FIG. 3.

FIG. 5 is a front view of the knee joint rehabilitation device of FIG. 3.

FIG. 6 is a perspective view of the knee joint rehabilitation device of FIG. 3 with a lower leg support shown in a stowed position.

FIG. 7 is a perspective view of another embodiment of a knee joint rehabilitation device including a lower leg support and including a goniometer or angle guide.

FIG. 8 is a closeup view of the knee joint rehabilitation device shown in FIG. 7 showing an angle guide or goniometer.

FIG. 9 is another closeup view of the knee joint rehabilitation device shown in FIG. 7 showing the angle guide or goniometer in perspective.

FIG. 10 is a picture of another embodiment of a knee joint rehabilitation device.

FIG. 11 is a picture of a closeup view of an angle guide or goniometer included on the knee joint rehabilitation device of FIG. 10

FIG. 12 illustrates how a knee may be placed on a support bar including an arrangement of pads for providing a first effective diameter for the support bar.

FIG. 13 illustrates how a knee may be placed on a support bar including an arrangement of pads for providing a second effective diameter for the support bar.

FIG. 14 illustrates how a knee may be placed on a support bar including an arrangement of pads for providing a third effective diameter for the support bar.

FIG. 15 illustrates how a knee may be placed on a support bar including an arrangement of pads for providing a fourth effective diameter for the support bar.

FIG. 16 is a flowchart of one embodiment of a method for knee joint rehabilitation using a knee joint flexion device or apparatus.

FIG. 17 is a flowchart of another embodiment of a method for knee joint rehabilitation using a knee joint flexion device or apparatus wherein a patient manually controls the position of a lower leg support during a therapeutic time period.

FIG. 18 is a picture of an individual using a knee joint flexion device or apparatus, the apparatus shown in a first position. As shown in FIG. 18, the knee joint flexion apparatus may include a handheld control for controlling the position of a lower leg support.

FIG. 19 is a flowchart of another embodiment of a method for knee joint rehabilitation using a knee joint flexion device or apparatus wherein a position of a lower leg support is adjusted during a therapeutic time period.

FIG. 20 is a picture of an individual using a knee flexion therapeutic apparatus showing a second position of the knee joint flexion device or apparatus shown in FIG. 18.

FIG. 21 shows a knee joint flexion device or apparatus included as part of a knee joint rehabilitation system.

FIG. 22 shows an embodiment of a support bar having a first effective diameter coupled to a height-adjustment bar.

FIG. 23 shows an embodiment of a support bar having a second effective diameter coupled to a height-adjustment bar.

DETAILED DESCRIPTION

The following terms as used herein should be understood to have the indicated meanings unless the context requires otherwise.

When an item is introduced by “a” or “an,” it should be understood to mean one or more of that item.

“Comprises” means includes but is not limited to.

“Comprising” means including but not limited to.

The term “effective outer diameter” of a support bar refers to the diameter of the support bar when considering any padding or other material around the component. For example, the effective outer diameter of a support bar includes the diameter of the support bar (e.g., the width of material through the center of the bar) adjusted for the thickness of one or more pads disposed about the support bar.

“Having” means including but not limited to.

A “hinge” means a joint that allows rotation of at least one of the components connected by the hinge with respect to the other.

A “joint” may include any connection that permits movement of the components connected by the joint with respect to one another. For example, a joint may provide for rotational, hinged, or pivotable motion between the components connected to one another by the joint. By way of example, a joint may include one or more pin connections, rod and sleeve connections, ball and socket connections, or the like.

Disclosed herein are knee joint rehabilitation systems and methods adapted for knee joint rehabilitation following knee joint surgery or other type of knee joint treatments, such as knee joint replacement, replacement or repair of any torn or damaged knee ligaments, bone fracture and bone fracture

surgery, internal fixation surgery, meniscus repair, treatment of arthritic conditions, knee injury recovery, regaining normal knee joint flexibility, and range of motion and strength recovery. The systems and methods may be used for reducing muscle spasm, pain, and swelling. The systems and methods may involve use of devices configured to provide and track appropriate flexion and extension support of the posterior intersection concerning the femur and tibia of an non-operative, arthritic, injured, fractured, or post-operative recovering knee joint, and to promote deliberate gravity assisted passive flexion and extension towards gradual knee joint muscle and tendon stretching, and mechanical decompression of the knee joint to alleviate pain, pressure, swelling, and general discomfort.

A knee joint rehabilitation system may comprise a knee joint flexion device or apparatus including a support bar configured for supporting the patient's knee and promoting bending of the patient's knee about a controlled angle when the knee is positioned over the support bar and the patient's leg is allowed to hang over the support bar to facilitate passive gravity-assisted stretching of the knee joint. The knee joint flexion device may include a system of pads configured for nesting inside one another and mounting to a support bar to adjust the effective diameter of the support bar and control the angle about which a patient's knee may bend. Alternatively, a plurality of support bars and/or pads may be used to provide different support bar effective diameters. Adjustment of the effective diameter of the support bar may facilitate incremental knee flexion during an individual's progression through a physical therapy regimen. Physical therapy regimens and methods making use of support bars with different effective diameters are described more fully herein including in reference to FIGS. 16, 17, and 19, for example. Some of the pads which may be used to provide different effective diameters of support bars and which may be included in or used with the knee flexion devices and methods described herein are also disclosed in U.S. Pat. No. 9,962,311 entitled “Method and Apparatus for Knee Joint Flexibility Rehabilitation,” and in US Patent Application Publication No. 20180169466 entitled “Method and Apparatus for Variable Knee Flexion Support,” the entire disclosures of which are incorporated herein by reference.

For example, FIG. 1 shows one embodiment of a knee joint flexion device 10. The knee joint flexion device 10 may be provided to a user in any of various forms of assembly. For example, the knee joint flexion device 10 may be provided in a fully assembled form or with one or more of the following components in a disassembled form and including instructions for assembly. The knee joint flexion device 10 includes a support bar 12 and a system of pads 14 useful for supporting a patient's knee. The system of pads 14 may, for example, include a plurality of pads configured for nesting inside one another and mounting to the support bar 12 to adjust the effective diameter of the support bar. The knee joint flexion device 10 may further include a tubular strut 16 having a first support foot 18 and a bracket 20 attached thereto. The height-adjustment bar 22 may be translatable or rotatably disposed in the tubular strut 16. For example, the height-adjustment bar 22 may fit into an aperture formed in the tubular strut 16 so that the height-adjustment bar may translate vertically within the tubular strut. The height-adjustment bar 22 may include a plurality of apertures 24 extending through the cross-section of the height-adjustment bar. A pin 26 may be affixed to the tubular strut 16 and configured for coupling with a selected one of the plurality of apertures 24 to secure the height-

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adjustment bar **22** in a selected position. The support bar **12** may be connected to the height-adjustment bar. For example, the support bar **12** may be fixedly disposed substantially perpendicularly to the height-adjustment bar **22**. In some embodiments, the support bar **12** may be removably coupled to the height-adjustment bar. One or more pads among the system of pads **14** may be disposed thereon to provide a given effective diameter among a group of available effective diameters and to receive a knee joint posterior more comfortably, as shown in FIG. **2**, for example. To vary the effective diameter or incrementally change the effective diameter of the support bar, one or more of the pads may be added or removed. A support strut **28** may be pivotably connected to the bracket **20**, the support strut **28** having a second support foot **30** attached thereto. The support strut **28** may be pivotably connected to the bracket **20** so that it may be positioned away from parallel to the tubular strut **16** to form an angle thereto and to provide stable support for device operation.

Another embodiment of a knee joint flexion device **100** is shown in FIGS. **3-6**. As shown therein, the knee joint flexion device **100** may include the height-adjustment bar **22**. The height-adjustment bar may again be translatably or rotatably and rotatably disposed in the tubular strut **16**, as similarly described for the flexion device **10**. Tubular strut **16** may, for example, be supported using a support foot **18** and support strut **28**. In other embodiments, the height-adjustment bar **22** may be adjustably supported in a vertical orientation in a different way. The flexion device **100** may comprise a lower leg support **32**. Lower leg support **32** may help to support a patient's leg when the patient's leg is extended over the support bar **12**, thereby reducing pressure upon anatomical structures of the patient's knee, including, for example, anatomical structures adjacent to the popliteal fossa of the patient's knee. The lower leg support **32** may, for example, be connected to the height-adjustment bar **22** at a joint **44** so that the position of the lower leg support may be adjustable. The joint **44** may, for example, include a pivot **41**. The pivot **41** may, for example, be positioned adjacent but slightly offset from the support bar **12**. In some embodiments, as shown in FIG. **18**, a knee joint flexion device may be provided with a controller **60** configured for use with one or more actuators **50**, **52**. The controller **60** may, for example, comprise a device sized and/or shaped for hand-held grasping with one or more buttons, knobs, triggers, switches, or other attachments disposed thereon for controlling extension and retraction of the one or more actuators **50**, **52**. Extension or retraction of the one or more actuators **50**, **52** may be used to rotate the lower leg support **32** about a hinge or other joint so as to adjust the relative position or angle of the lower leg support **32**. For example, extension or retraction of the one or more actuators **50**, **52** may be used to rotate the lower leg support **32** about a hinge or other joint so as to provide for adjustment of a relative angle between the lower leg support **32** and the height-adjustment bar **22**. Thus, in some embodiments, a position of the lower leg support **32** with respect to the height-adjustment bar **22** may be controlled to provide for stronger or weaker engagement with the patient's leg and to support more or less of a weight of the patient's leg. For example, in some embodiments, lower leg support **32** may be controlled to ease a patient between stages in a therapeutic regimen. For example, more or less of the weight of the patient's leg may be supported by the lower leg support **32** when an individual is transitioning from one stage in a therapeutic regimen associated with a first effective diameter of the support bar **12** to a next

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stage in the therapeutic regimen associated with a second effective diameter of the support bar.

In some embodiments, as shown in FIGS. **3** and **4**, the lower leg support **32** may comprise a shaped pack or panel (e.g., a panel with associated padding disposed thereon) including a first recess or groove **34** for accepting a posterior portion of the patient's leg. The lower leg support **32** may further include a notch or opening **33** formed therein. The recess or groove **34** may be formed within a gap between a left-side wall or ridge **36** and a right-side wall or ridge **38** of the lower leg support **32**. Lower leg support **32** further includes a first end **40** and a second end **42** distal from the first end. In some embodiments, the lower leg support **32** may generally increase in thickness along at least along a portion of the length of the lower leg support **32** running from the first end **40** to the second end **42**. In this configuration, a second recess or groove **48** may be formed adjacent to the first end **40** of the lower leg support **32** in the space between the padded support bar **12** and the lower leg support **32**. The second recess or groove **48** may run about perpendicular to the first recess or groove **34**. The lower leg support **32** may, for example, comprise a top surface **75** contoured to comfortably receive the patient's calf muscle.

The configuration of the lower leg support **32** (e.g., the surface shape, size, and position of the grooves **34**, **48**) may help to support the patient's lower leg more fully and allow the patient's leg to be more comfortably received within the grooves as the lower leg support is moved through different angular positions. In some embodiments, the shape of the lower leg support **32** and positioning of one or more force or pressure sensors **70a**, **70b**, and **70c** disposed thereon may help to provide reliable contact between the patient's leg and at least one of the sensors **70a**, **70b**, and **70c** so that reliable sensor signals may be obtained for different angular positions of the lower leg support **32**. To that point, lower leg support **32** may be connected to the height-adjustment bar **22** using the joint **44**, wherein the joint may be positioned on the height-adjustment bar **22** and spaced apart from the support bar **12**. The joint **44** may comprise the pivot **41**. The relative position between the joint **44** and the support bar **12** may be useful because it allows a user to add or take away one or more pads from the system of pads **14** more easily (e.g., without having the connecting joint **44** interfere with addition or removal of pads from the support bar **12**). For example, in some embodiments, the joint **44** may be offset from the support bar **12** by about 3 inches to about 6 inches or by some other suitable distance. However, with positioning of the joint **44** and associated pivot **41** at a distance below the support bar **12**, the pivot is also spaced significantly apart from the knee joint of a patient when the patient's knee is supported by the support bar **12**. In this geometry, the resting position of the patient's leg on atop surface **75** of the lower leg support **32** becomes dependent upon the flexion angle. In other words, because the patient's leg and the lower leg support **32** are rotated about different pivots (e.g., the pivot of the patient's knee and pivot **41** of the hinge **44**) that are spaced apart from each other, the patient's leg may slide along the top surface **75** of the lower leg support when it is adjusted.

For example, as shown in reference to FIGS. **18** and **20**, for some patients, including some patients with significant calf musculature, a patient's calf may seat significantly within the groove **48**, particularly at higher angles of flexion. However, at lesser angles of flexion, the patient's leg may be urged slightly forward so that the patient's leg is seated more fully in the groove **34**. In the embodiment shown in FIGS. **3-6**, the shape of the lower leg support **32** in combination

with the positioning of forward-facing sensors **70a**, **70b** and more rearward facing sensors **70c** helps to comfortably support the patient's leg and helps to guarantee that the patient's leg is in proper contact with at least one sensor throughout different flexion angles. In some embodiments, a lower leg support **32** may be provided with two grooves **34**, **48** wherein at least one force or pressure sensor **70a**, **70b**, is generally positioned in the first groove **34** and at least one other force or pressure sensor **70c** is positioned in the second groove **48**.

The first end **40** of the lower leg support **32** may be connected to the joint **44** so that the lower leg support may be hingedly or pivotably disposed with respect to the height-adjustment bar **22**. The joint **44** may, for example, comprise one or more pin connections, a rod and sleeve connection, or a ball and socket connection. In some embodiments, the joint **44** may comprise a plate **58** that is secured (e.g., bolted, screwed, welded, or secured using other suitable means) to the height-adjustment bar **22**. For example, the plate **58** may be secured to the height-adjustment bar **22** using one or more bolts or screws or using a weldment. The plate **58** may comprise a first leaf of a pair of leaves of the joint **44** (which is a hinge in this example) through which a knuckle is formed. A pin may extend through the knuckle so that a second leaf among the pair of leaves may rotate about the pin (e.g., so that it pivots away from the plate **58**). The second leaf of the hinge may, for example, be secured to the first end **40** of the lower leg support **32** so that the lower leg support **32** moves about the joint **44**.

As described above, in some embodiments, a position of the lower leg support **32** may be changed using one or more actuators **50**, **52**, such as may be controlled using a controller **60** (shown in FIG. **18**, for example). In some embodiments, one or more of the actuators **50**, **52** may be configured for receiving a signal from the controller **60** to initiate actuation thereof and adjust the lower leg support **32**. For example, the actuators **50**, **52** may be linked to the controller **60** using a wired or wireless connection and may include components (e.g., a processing unit and related circuitry) for processing commands sent therebetween. In some embodiments, the control system may include a unit of memory which may be useful in providing a capability to store one or more programs including one or more routines suitable for controlling the lower leg support **32** in a predetermined or preselected way. In some embodiments, a controller may run a program or application useful for controlling the position of the lower leg support **32**. Such an application may, for example, be downloaded or updated from a website. In some embodiments, the actuators **50**, **52** may be controlled using a separate standalone computer, such as may directly control the actuators **50**, **52** or control actuation thereof through the controller **60**. The controller **60** may comprise a single button or switch adjustable between various settings, including, a first setting for raising the lower leg support and a second setting for lowering the lower leg support. In some embodiments, the controller **60** may be configured for receiving a preselected input such as a digital setting of an angular position between the lower leg support **32** and the height-adjustment bar **22**. In another embodiment, the controller **60** may be equipped with a rotating knob such as may be turned to adjust a position for the lower leg support **32**. A computer may, for example, be configured for running a program or application useful for controlling the position of the lower leg support **32**.

In the illustrated embodiment, the lower leg support **32** may be connected to a pair of actuators **50**, **52**. The actuators **50**, **52** may, for example, comprise a cylinder barrel and an

associated piston rod extendable and retractable therefrom. The actuators **50**, **52** may be pneumatic, hydraulic, mechanical, electrical, or of another suitable type. The actuators **50**, **52** may, for example, operate pneumatically, hydraulically, electrically, mechanically, or using another suitable source of energy. Actuator **50** (which may also be referred to as the left-side actuator) is shown in FIG. **4**. Actuator **52** (which may also be referred to as the right-side actuator) is disposed behind the actuator **50** so that it is not shown in FIG. **4**. Right-side actuator **52** is shown in the perspective view of FIG. **3**. Both the left-side and right-side actuators **50**, **52** can be viewed together in FIG. **5**, which shows a front view of the knee joint flexion device **100**. Each of the actuators **50**, **52** includes a first end and a second end. With reference to FIG. **4**, left-side actuator **50** includes first end **53** and second end **55**. Left-side actuator **50** is pivotably connected to the lower leg support **32** via the joint **56** at first end **53**. Right-side actuator **52** is similarly connected to the lower leg support **32** via the joint **51**. The actuators **50**, **52** may further be connected to the height-adjustment bar **22** though a connecting bracket **59**. Connecting bracket **59** may be secured to the height-adjustment bar **22** using any suitable structure or means such as a screw, bolt, or weldment for example. Right-side actuator **52** may be pivotably connected to the height-adjustment bar **22** through the connecting bracket **59** at joint **57**. Left-side actuator **50** may be pivotably connected to the height-adjustment bar **22** through the connecting bracket **59** at joint **54**.

As shown in FIG. **3**, in some embodiments, the knee joint flexion device **100** may include one or more sensors. For example, the sensors **70a**, **70b**, **70c** may be disposed on the lower leg support **32** and configured for collecting force or pressure data useful for guiding and/or tracking patient progress and compliance with a prescribed therapy regimen. One or more additional sensors **71** may further be provided as may be used, for example, in detection of contact between a patient's knee and the support bar **12**. For example, in the embodiment shown in FIGS. **3-6**, a pair of sensors **71** are included on an outermost pad mounted on the support bar. Methods of guiding and/or tracking patient progress and compliance with a prescribed therapy regimen which, may, in some embodiments, make use of one or more of the sensors **70a**, **70b**, **70c**, and **71** are further described herein, including, in relation to method **240** shown in FIG. **19**, for example. In some embodiments, one or more of the sensors **70a**, **70b**, **70c**, and **71** may comprise a unit of memory configured for storing collected sensor data. The stored data may, for example, be accessed when the data is downloaded or otherwise accessed for review. In some embodiments, one or more of the sensors **70a**, **70b**, **70c**, and **71** may comprise a transmitter or transceiver so that information may be transmitted from the sensors to one or more of controller **60** or a separate computer system **90**. In some of those embodiments, data may be collected and used in real time in one or more routines used for controlling a position of the lower leg support **32**. For example, as described in methods **200** and **240** (shown in FIGS. **16** and **19**), in some therapeutic procedures described herein, a level of engagement or disengagement of the lower leg support **32** from a patient's leg may be determined using collected sensor data, which may, in some embodiments, be then used in setting or adjusting a position of the lower leg support.

Another embodiment of a knee joint flexion device **150** is shown in FIGS. **7-9**. As shown therein, the knee joint flexion device **150** may comprise an angle guide or goniometer **66**. The goniometer **66** may, for example, be used to collect data facilitating reproducible adjustment of the lower leg support

32 and/or used for recording of the position of the lower leg support position during a therapeutic regimen. With reference to FIGS. 8 and 9, the angle guide or goniometer 66 may comprise a rod 72 and a sleeve 74, such as may be slidably engaged with each other. The rod 72 may comprise markers 76 in the form of notches, for example. A user may manually read the position of the sleeve 74 against the markers 76 to record a relative position between the rod 72 and the sleeve. This measurement may be correlated with a given angle between the lower leg support 32 and the height-adjustment bar 22. For example, in one embodiment, the markers 76 may be calibrated so that they directly indicate a relative angle between the lower leg support 32 and the height-adjustment bar 22.

In some embodiments, a knee joint flexion device may be configured for automatic adjustment of the lower leg support 32 to a desired position. This may, for example, make use of data collected with the goniometer 66 or with another suitable goniometer described herein. For example, one or more goniometers may be disposed at one or more joints between any two of the lower leg support, height-adjustment bar, and at least one of the actuators 50, 52. In one example, goniometer 66 may be configured to automatically record the relative position of the lower leg support 32 through an electromechanical coupling between the rod 72 and sleeve 74. In one such embodiment, the rod 72 may include one or more sensors disposed therein configured for detection of a magnetic field. For example, the rod 72 may include one or more hall sensors or other components configured for detection of a magnetic field. The sleeve 74 may include a magnet so that when the sleeve 74 and rod 72 move against each other, changes in relative position between the sleeve and rod may be detected using the sensors. Alternatively, one or more goniometers may be coupled to one or more of the joints 44, 54, 56. For example, an electromechanical goniometer disposed at one or more of the joints 44, 54, 56 may comprise a potentiometer rotationally mounted therein. The goniometers may provide a signal comprising data for the angular position of the lower leg support 32. In some embodiments, the actuators 50, 52 may be configured to receive a signal indicative of an angular position so that the lower leg support 32 may be automatically adjusted to a selected angular position, such as a position previously recorded in a previous session of a therapeutic regimen. This may be executed through data transmission involving the actuators 50, 52 and one or more of controller 60 and a standalone computer, for example.

Another embodiment of an angle guide or goniometer 80 is shown in FIGS. 10 and 11. As shown therein, in some embodiments, knee joint flexion device 100 may include an angle guide or goniometer 80. Goniometer 80 may include markings 82 allowing a user to manually record a position (e.g., a position calibrated in degrees or relative units with respect to a height-adjustment bar) of the lower leg support 32. Alternatively, an electromechanical type of goniometer may be included in a knee joint flexion device.

Referring again to FIG. 2 and with further reference to FIGS. 12-15, a knee joint flexion device 10, 100, 150 as described herein may be used in a knee rehabilitation therapy regimen. When an individual is using a knee joint flexion device 10, 100, 150 the support bar 12 may be placed at the popliteal fossa of the knee. Such placement naturally results in the application of pressure to sensitive biological structures that are located in or traverse the fossa. Some such structures are, but not limited to, the tibial nerve, popliteal nerve, popliteal artery, small saphenous vein, common peroneal nerve, popliteal lymph nodes, and Dharmy vessels.

This pressure may be attenuated using a pad or system of pads 14 as described above. The system of pads 14 may help to distribute and reduce pressure on these structures as compared, for example, to use of an exposed hard support bar, such as a metal or plastic bar. Accordingly, the time a particular pad of a therapeutic regime may be reasonably used may be increased.

Placement of the support bar 12 at the popliteal fossa permits the knee joint to "expand" to various degrees depending on the effective diameter of the support bar when a patient hangs the knee over the support bar. Such joint expansion beneficially promotes knee joint flexibility, particularly when the knee joint is progressively treated using pads of different diameter as described herein. For example, as the strength and flexibility of a knee joint for a given patient improves the patient may choose to transition from a first level of therapy associated with a first effective diameter of the support bar to a second level of therapy associated with a second effective diameter of the support bar.

By varying the effective diameter of the support bar 12 through various support bar diameters or by addition or removal of pad layers from the system of pads 14 a physical therapist (or patient, or doctor or other caregiver, as the case may be) may allow therapy to progress through each layer or pad support size based on the patient's therapeutic requirements throughout the course of a series of therapy sessions constituting a knee rehabilitation therapy regimen, thus better allowing for a patient-specific recovery regimen. Such requirements may include the level of flexion intended for a particular session based on patient comfort level and therapy goals for the session. Other requirements or needs defined in a therapeutic regimen may, for example, include a level of support provided by a lower leg support, time spent in a routine wherein the patient's leg was supported by a lower leg support, and a time spent at a given level of knee flexion. A system may provide a plurality of pads or support bar diameters having dimensions suitable for allowing various increments between ranges of motion. An embodiment suitable for treating a wide range of knee therapy patients comprises a set of four nested pads; however, the number of pads to be included may vary according to the treatment regimen and patient condition. The number of pads should allow for a variation of effective diameter of the support bar as added or subtracted and thus result in the therapy regime allowing for a variation in range of motion at the knee as the lower leg is rotated about the support bar. Some patients may require starting physical therapy at a larger (or much smaller) effective diameter than other patients, and so the number of pads may be adjusted accordingly. Furthermore, the number of pads may be inversely proportional to their thickness; that is, fewer thick pads may be used where larger range-of-motion increments may be beneficially used, or more thin pads may be used where smaller range-of-motion increments may be beneficially used.

In various embodiments, the pad layers 6, 7, 8, and 9 may incrementally increase the diameter of the support bar 12 to allow a physical therapist a method of incrementally increasing a patient's knee flexion. Although four pad diameters are shown in the exemplary starting configuration of pads for a therapeutic routine shown in FIG. 12, any suitable number or range of diameters may be used. For example, a post-operative patient may have limited range of motion in a knee 13, which may allow only a relatively large angle between the femur 15 and tibia 17. For such a patient, a plurality of layers may be used to increase the effective diameter of the support bar 12 so as to better support a knee during the first

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sessions of a knee therapy regimen wherein each layer of pad may correspond to an angle of flexion α at the knee joint to be achieved between the upper and lower segments of limb. For example, a first pad layer 6 may correspond to a flexion of approximately 90° , a second pad layer 7 to a flexion of approximately 105° , a third pad layer 8 to a flexion of approximately 120° , a fourth pad layer 9 to a flexion of approximately 135° , and so forth. Thus, in some embodiments, pad thickness and diameter may provide angles of flexion in 15° increments; however, other increments may be used. For example, a patient with flexion of only 80° may use the first pad layer 6 to achieve flexion of 90° , and then transition to use of the second pad layer 7 to achieve flexion of approximately 105° .

As may be seen in the embodiment of FIG. 12, a first pad layer 6, a second pad layer 7, a third pad layer 8, and fourth pad layer 9 are disposed around the support bar 12, thus providing a larger-diameter support surface. While a patient is supine one of the patient's legs may be positioned over the support bar 12 such that the support bar is disposed substantially behind the patient's knee 13 at the popliteal area. The patient's leg may then be hung over the support bar. In some embodiments, at least a portion of the weight of the patient's leg may be supported by lower leg support 32. The patient's lower leg, generally including the tibia 17 and foot 19, extends beyond the support bar 12. The support bar 12 thus serves as a fulcrum point 21 or area upon which the knee joint region of the leg may rest and the portion of the lower leg may pivot. The larger diameter support bar ensures that the support bar 12 provides a fulcrum point 21 farther away from the knee, thus allowing the knee joint to expand. With the leg disposed on the support bar 12, the physical therapist may allow gravity to pull the lower leg downward through a range of motion to help stretch the knee joint tissues. The lower leg may effectively serve as a lever, with the femur and patient's body weight at the knee side of the fulcrum point serving as an anchor weight. The gravity force serves to gently distract the knee joint, thus allowing the knee joint to separate and stretch substantially without creating compressive forces in the joint. The fulcrum placement may alter the mechanical advantage of the lever so as to selectively multiply the effect of gravity as the physical therapist manipulates the lower leg. A larger diameter may allow a physical therapist to treat a knee joint shortly after surgery, for example, and may allow for increased patient comfort even with substantial post-operative swelling. A larger diameter may effectively provide a larger surface area on which the patient's leg may rest, thus spreading the fulcrum pressures and correspondingly reducing patient discomfort. For self-applied regimes, a patient or patient caregiver may allow gravity to pull the lower leg downward through a range of motion to help stretch the knee joint tissues.

Voluntary knee contraction while the leg is so supported may be further applied to assist in strengthening the knee. In some therapy regimes, passive gravity-assisted stretching may be alternated with voluntary knee contraction. Further, in some embodiments, a lower leg support 32 may sometimes be used to support at least a portion of the weight of the patient's leg. The lower leg support may, for example, be used progressively less over the course of a therapy so that a decreasing proportion of the weight of the patient's leg is supported thereby.

As the range of motion or flexibility and strength of the knee 13 increases over the course of the therapy regime, the fourth pad layer 9 may be removed from the support bar 12, thus reducing the effective diameter of the support bar, as

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may be seen in the embodiment of FIG. 13. For example, in some embodiments, strength in the knee 13 may be determined when the patient achieves a level of knee flexion without having any significant proportion of the patient's weight being supported using the lower leg support 32. The third pad layer 8 may thus support the knee 13 directly. The smaller effective diameter of the third pad layer 8 may allow for a more acute angle between the femur 15 and tibia 17 than the angle allowed by the fourth pad layer 9. The third pad layer 8 may allow the fulcrum point 21 to be closer to the knee joint, thus changing the mechanical advantage provided by the support bar and further multiplying the force generated by gravity or by physical therapist pressure 11.

As the range of motion of the knee 13 further increases over the course of the therapy regime, the third pad layer 8 may be removed from the support bar 12, thus reducing the effective diameter of the support bar, as may be seen in the embodiment of FIG. 14. The second pad layer 7 may thus support the knee 13 directly. The smaller effective diameter of the second pad layer 7 may allow for a more acute angle between the femur 15 and tibia 17 than the angle allowed by the third pad layer 8. The second pad layer 7 may allow the fulcrum point 21 to be closer to the knee joint, thus changing the mechanical advantage provided by the support bar and further multiplying the force generated by gravity or by physical therapist pressure 11.

As the range of motion of the knee 13 still further increases over the course of the physical therapy regime, the second pad layer 7 may be removed from the support bar 12, thus reducing the effective diameter of the support bar, as may be seen in the embodiment of FIG. 15. The first pad layer 6 may thus support the knee 13 directly. The smaller effective diameter of the support bar 12 may allow for an even more acute angle between the femur 15 and tibia 17 than the angle allowed by the second pad layer 7. The first pad layer 6 may allow the fulcrum point 21 to be closer to the knee joint, thus changing the mechanical advantage provided by the support bar and yet further multiplying the force generated by gravity or by physical therapist pressure 11. Thus, by changing the effective diameter of the support bar, the physical therapist may effectively control the maximum flexion angle.

To transition between different levels of therapy, the patient may, for example, disengage from use of the device to allow for removal of the outermost pad mounted to the support bar. Once the outermost pad is removed (reducing the effective diameter of the support bar 12), the patient may then reposition their knee upon the support bar to complete transition from one level of therapy to another level of therapy. This may, for example, be done in two different therapeutic sessions. However, at least in some situations, a patient may transition between levels of therapy in a single therapeutic session. For example, following treatment using a support bar of one effective diameter a patient may get up and walk before returning to engage the support bar at a reduced effective diameter. However, in some situations, it may be useful to transition a patient between levels of therapy without having the patient fully disengaging from the device.

For example, in some embodiments, a lower leg support 32 may sometimes be used to assist in transitioning a patient between different levels of therapy. For example, a patient may be treated with passive gravity-assisted stretching using a first effective diameter as described above. To transition to a next level of therapy. The lower leg support 32 may be moved to support at least a portion of the weight of the patient's leg. With at least a portion of the weight of the

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patient's leg supported, a caregiver may then slide an outermost pad out of engagement with the patient's knee. In this operation, the lower leg support 32 may function to help support the patient's leg during transition between the two levels of therapy. In this way, transition between levels of therapy may be made more comfortably without having the patient completely disengage from the device. Alternatively, the patient may fully disengage from a device, but when they first reengage the device at a new level of flexion, a lower leg support 32 may be used to support a weight of the patient's leg. In either case, at least the patient's first encounter with a new therapeutic level may make use of the lower leg support 32. The position of the lower leg support 32 may then be adjusted to slowly ease the patient's knee into proper engagement with the support bar. For example, a patient or caregiver may slowly adjust the angular position of the lower leg support 32 so that the lower leg support slowly disengages fully or in part from the patient's leg thereby gently transitioning the patient to the next level of therapy.

In some embodiments, disengagement of the patient's leg with the lower leg support 32 may be execute using a predetermined routine. For example, the one or more actuators 50, 52 may receive a control signal so that the angle between the lower leg support 32 and the height-adjustment bar 22 may change at a predetermined rate or following some predetermined pattern defined in a routine. For example, in one routine, the lower leg support 32 may disengage from supporting the patient's leg by changing its angular position at a rate, as may be expressed in degrees per minute. In some embodiments, a rate of disengagement between the lower leg support 32 may be controlled using feedback from sensor data. For example, the lower leg support 32 may disengage from the patient's leg slowly so that the weight of the patient's leg supported thereby changes in a controlled manner based on a signal received by one or more of the sensors 70. A rate of disengagement may then be calibrated or measured in units of pounds of force per minute, for example. Disengagement of the lower leg support 32 from the patient's leg may sometimes be set to provide only partial disengagement from the patient's leg. For example, the lower leg support 32 may disengage from the patient until a threshold pressure or force is measured by one or more of the sensors 70.

FIG. 16 shows an embodiment of a method 200 for rehabilitation of a patient's knee joint using a knee joint flexion device or apparatus. As shown in step 202, a pad or set of pads useful for providing a desired effective diameter of a support bar 12 may be selected. For example, in some therapy sessions, one or more pads for achieving an effective diameter of a support bar may be prescribed by a physical therapist, doctor, or other qualified individual as may be appropriate for a patient at a particular stage in a therapeutic regimen. At step 204, a support bar 12 (e.g., a support bar having an appropriate diameter pad or number of nested pads to achieve the intended certain effective diameter) may be positioned under the knee. The support bar 12 may be disposed at the popliteal area of a knee joint while the patient is lying on their back in a supine position. The support bar 12 may then be adjusted as may be needed to provide a height to permit the lower leg to hang over the support bar such that a patient's foot does not rest on the surface upon which the therapy session is being performed. In some embodiments, lower leg support 32 may further be adjusted to a position so that it is ready to receive a leg of the patient when the patient's knee is hung over the support bar 12.

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At step 206, the patient's knee may be hung over the support bar 12. This may, for example, be accomplished by the patient with or without the assistance of a caregiver. The lower leg support 32 may be used to gradually ease the patient into engagement with the support bar 12. For example, in some embodiments, the patient's knee may be hung over the support bar 12 and allowed to immerse into one or more pads among the system of pads 14. Once the patient's knee has, fully or at least in part, settled onto the padded support bar 12, the lower leg support 32 may be brought into engagement with the patient's leg. For example, in some embodiments, it may be useful for a caregiver to guide the patient's knee so that it is properly seated on the support bar 12 in a suitable position for gravity assisted flexion before the lower leg support 32 is moved into engagement. This initial positioning may involve allowing the patient's knee to at least in part become immersed into the padded support bar 12. Once this position is established, the lower leg support 32 may be moved into a position for engagement with the lower part of the patient's leg. A caregiver may help to support or guide the patient's leg during initial engagement of the patient's knee with the support bar 12 as well as during initial engagement of the patient's lower leg with the lower leg support 32. The caregiver may further take care to ensure that engagement of the patient's leg with the lower leg support 32 does not inadvertently cause the patient to shift the position of the patient's knee on the support bar so that it is no longer in a desired position for passive gravity assisted flexion.

In some embodiments, engagement between the lower leg support 32 and the patient's leg may generally be made together with or at about the same time as when the patient's knee is hung over the support bar 12. Thus, advantageously, the lower leg support may be used to help relieve stress on the patient's knee during an initial time period when the patient is first engaging with the device and when the patient may experience discomfort with the procedure. An initial level of engagement between the patient's leg and the lower leg support 32 may be established in various ways. For example, in some embodiments, the lower leg support 32 may be adjusted until visual identification is made of engagement between the posterior side of the patient's lower leg and the lower leg support. In other examples, the lower leg support 32 may be brought to a preselected initial position (e.g., a position established or read using a goniometer or angle guide), such as may be specified by a doctor or other caregiver. In other embodiments, the lower leg support 32 may be adjusted to an initial position established using collected sensor data provided from one or more of the sensors 70a, 70b, and 70c.

Still referring to step 206, the lower leg support 32 may be moved from its initial position to a second position wherein it is substantially disengaged from the patient's leg. For example, in one embodiment, the lower leg support 32 may be adjusted from an engaged to a fully disengaged position over a first time period. Thus, the patient's leg may be significantly supported by the lower leg support 32 during the initial setup or first time period described above. However, during a second time, describe below in relation to step 208, the patient's leg may flex under passive gravity assisted conditions without being significantly support by the lower leg support 32. In one example, the patient may actively control the position of the lower leg support 32 during the first time period. For example, as shown in FIG. 18, the patient may use a handheld controller to slowly disengage the lower leg support 32 from the treated leg. In another example, the lower leg support 32 may be controlled to

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automatically disengage from the patient's leg. For example, one or more of the actuators **50**, **52** may be controlled using a predetermined routine suitable for slowly disengaging the lower leg support **32** from the patient's leg. In some of those embodiments, disengagement of the lower leg support **32** from the patient's leg may be guided or controlled using sensor data collected using one or more of the sensors **70a**, **70b**, and **70c**.

At step **208**, the patient may allow their knee to flex under passive gravity assisted conditions for a second period of time. In some regimens, a caregiver may further prescribe that the patient engage in some other exercise during the second time period, such as leg extensions, for example. In the step **210**, the patient may disengage from the support bar. For some patient's, disengagement from the support bar may include adjusting the lower leg support **32** to reduce pressure on the patient's knee. For example, the lower leg support **32** may be engaged or reengaged with the patient's leg to take pressure off the knee joint. This may, for example, make it easier for a patient to dismount a knee joint flexion device. Alternatively, the lower leg support **32** may be moved to a position wherein it may help to support the patient's leg making it easier for a caregiver to exchange, add, or remove one or more pads from the support bar **12**. For example, the lower leg support **32** may be adjusted to a position wherein a physical therapist may more easily exchange one pad for another or remove a pad from a group of pads so as to continue treatment of the patient with a support bar **12** with a different effective diameter.

FIG. **17** shows another embodiment of a method **220** for rehabilitation of a patient's knee joint using a knee joint flexion device or apparatus. As similarly explained above in relation to method **200**, in step **222**, a pad or set of pads useful for providing a desired effective diameter for a support bar **12** may be selected. At step **224**, the support bar **12** may be positioned under the knee. The support bar **12** may be disposed at the popliteal area of a knee joint and adjusted as may be needed to provide a height to permit the lower leg to hang over the support bar such that a patient's foot does not rest on the surface upon which the therapy session is being performed. In some embodiments, lower leg support **32** may further be adjusted to a position so that it is ready to receive a leg of the patient when the patient's knee is hung over the support bar **12**.

At step **226**, the patient's knee may be hung over the support bar **12**. This may be accomplished by the patient alone or with assistance of a caregiver depending, for example, on the strength of the patient's knee. In some embodiments, the patient's knee may be hung over the support bar **12** and allowed to immerse into one or more pads among the system of pads **14**. Once the patient's knee has, fully or at least in part, settled onto the padded support bar **12**, the lower leg support **32** may be brought into engagement with the patient's leg. Thus, the patient's leg may hang over the support bar with at least a portion of the weight of the patient's leg being supported by the lower leg support **32**.

The patient's leg may be allowed to naturally extend under conditions of passive gravity assisted flexion. A portion of the weight of the patient's leg may be supported by the lower leg support. As shown in FIG. **18**, the patient may be provided with a controller **60** so that they may manually adjust the position of the lower leg support **32** to achieve a reasonable level of comfort as gravity works on the knee to slowly allow the knee joint to expand. In another embodiment, a knee joint flexion device may include a mechanical lever coupled to the lower leg support **32**. The

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mechanical lever may be sized in length so that a user may control the position of the lower leg support **32** from a supine position. Thus, in the embodiment of method **220**, the patient may control the lower leg support **32** actively during the therapeutic stage of gravity induced flexion. In some of those embodiments, data may be collected from one or more of the sensors **70a**, **70b**, **70c**, and **71** and made available for review by an attending doctor, physical therapist, or other caregiver. For example, a caregiver may receive data to understand when the patient may achieve a level of flexion without relying significantly on the lower leg support **32**. This may, for example, be evaluated as one indication that the patient may be ready for a next stage in flexion therapy.

As shown at step **228**, the patient may disengage from the support bar. For some patient's, disengagement from the support bar may include adjusting the lower leg support **32** to reduce pressure on the patient's knee to assist the patient in fully or partially disengaging from the device. For example, the support bar **12** may be raised to help support the patient's knee when the patient is dismounts to fully disengage from the device. In another example, the patient may only partially disengage from the device. For example, the lower leg support **32** may be adjusted to a position wherein a physical therapist may more easily exchange one pad for another or remove a pad from a group of pads to change the effective diameter of the support bar.

FIG. **19** shows another embodiment of a method **240** for rehabilitation of a patient's knee joint using a knee joint flexion device or apparatus. As shown in step **242**, a pad or set of pads useful for providing a desired effective diameter of a support bar **12** may be selected. At step **244**, a support bar **12** may be positioned under the knee and adjusted as may be needed to provide a height to permit the lower leg to hang over the support bar such that a patient's foot does not rest on the surface upon which the therapy session is being performed. Lower leg support **32** may further be adjusted to a position so that it is ready to receive a leg of the patient when the patient's knee is hung over the support bar **12**.

At step **246**, the patient's knee may be hung over the support bar **12**. For example, the patient's knee may be hung over the support bar **12** and allowed to immerse into the padded support bar **12**. Once the patient's knee has, at least in part, settled onto the padded support bar **12**, the lower leg support **32** may be brought into engagement with the patient's leg. Thus, the patient's leg may hang over the support bar **12** with at least a portion of the weight of the patient's leg being supported by the lower leg support **32**. In this embodiment, the position of the lower leg support **32** may be automatically adjusted during gravity assisted flexion. For example, in one embodiment, the lower leg support **32** may be set at an initial level of engagement with the patient's leg, such as may, for example, be accomplished when a user visually establishes engagement, by setting the support to a desired goniometer position, or using sensor data to provide an indication of desired level of support. At periodic intervals, the lower leg support **32** may then be automatically or manually adjusted. For example, the lower leg support **32** may be controlled using a preselected routine to slowly decrease a level of engagement with the patient's leg. In some embodiments, the lower leg support **32** may be controlled using a preselected routine to provide alternating levels of support. For example, during some times during a therapeutic session, the lower leg support **32** may be adjusted so as to slowly decrease a level of engagement or to fully disengage from the patient's leg. However, at other times the lower leg support **32** may be moved into a position

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wherein engagement with the patient's leg is greater (e.g., a greater proportion of the weight of the patient's leg may be supported by the lower leg support). Such embodiments may be useful because pressure on the patient's knee may be periodically reduced during a therapeutic session, which may allow some patient's to be treated for increased periods of time.

As shown at step **248**, the patient may disengage from the support bar. For some patient's, disengagement from the support bar may include adjusting the lower leg support **32** to reduce pressure on the patient's knee to assist the patient in fully or partially disengaging from the device. For example, the support bar **12** may be raised to help support the patient's knee when the patient is dismounts to fully disengage from the device. In another example, the patient may only partially disengage from the device. For example, the lower leg support **32** may be adjusted to a position wherein a physical therapist may more easily exchange one pad for another or remove a pad from a group of pads to change the effective diameter of the support bar.

In some embodiments, as shown in FIG. **21** a knee joint rehabilitation system may include a knee joint flexion device (such as the knee joint flexion device or apparatus **100** in the embodiment shown). The system **300** may further include a controller **60**, such as may be provided with a wired connection for coupling to the device **100**. The system may be provided with or otherwise used with a computer system **90**. The computer system may, for example, comprise a desktop computer programmed using one or more tangible computer media **92**. A controller including an integrated processor or separate computer may be configured for downloading of an application or program useful for controlling the position of a lower leg support **32** through actuation of one or more actuators.

A knee joint flexion device, system, and related methods are described. The devices, systems, and methods are described herein. In some of those embodiments, reference may be made to a patient, such as an individual who is recovering from a knee joint injury. However, the devices, systems, and methods may, in some embodiments, be used or executed by an individual using a device or system herein for maintaining flexibility of the knee joint.

As described herein, a knee joint flexion device **10**, **100**, **150** may include a system of pads configured for nesting inside one another and mounting to a support bar to adjust the effective diameter of the support bar and control the angle about which a patient's knee may bend. Alternatively, a plurality of support bars and/or height-adjustment bars may be used to provide different support bar effective diameters. For example, as shown in FIGS. **22** and **23** supports bars **120**, **220** with different effective diameters could be used as an alternative to a system of pads configured for nesting inside one another and mounting to a support bar to adjust the effective diameter of the support bar. The support bars **120**, **220** may, for example, be removably mounted to a height-adjustment bar so as to provide different effective diameters used in a treatment regimen. Alternatively, a plurality of different height-adjustment bars, each with a different support bar **120**, **220** connected thereto could be used to provided different effective diameters for use in a treatment regimen.

Although the disclosed subject matter and its advantages have been described in detail, it should be understood that various changes, substitutions and alterations can be made herein without departing from the subject matter as defined by the appended claims. Moreover, the scope of the present application is not intended to be limited to the particular

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embodiments of the process, machine, manufacture, composition, or matter, means, methods and steps described in the specification. Among other things, any feature described for one embodiment may be used in any other embodiment, and any feature described herein may be used independently or in combination with other features. For example, certain embodiments herein may describe use of sensors and angle guides (or goniometers). Unless the context clearly dictates otherwise, those components may be used in any of the different devices described herein. As one will readily appreciate from the disclosure, processes, machines, manufacture, compositions of matter, means, methods, or steps, presently existing or later to be developed that perform substantially the same function or achieve substantially the same result as the corresponding embodiments described herein may be utilized. Accordingly, the appended claims are intended to include within their scope such processes, machines, manufacture, compositions of matter, means, methods, systems or steps.

What is claimed is:

1. A knee joint rehabilitation system comprising:

a knee joint flexion device, comprising:

a support bar fixedly connected to a height-adjustment bar or configured for removable coupling to said height-adjustment bar, the support bar being further configured for receiving at least one of a plurality of pads so as to provide a padded support bar, the padded support bar being configured to receive a knee joint posterior;

said plurality of pads configured to removably mount to the support bar to adjust an effective outer diameter of the support bar;

a lower leg support connected to the height-adjustment bar using a first joint so as to allow for changes in an angular position between said lower leg support and said height-adjustment bar when the position of the lower leg support is adjusted;

a goniometer configured for indicating the angular position of said lower leg support; and

an actuator configured for extension and retraction so as to adjust the angular position between said lower leg support and said height-adjustment bar, said actuator including a first end and a second end, said actuator being connected to the lower leg support at said first end using a first hinge and connected to the height-adjustment bar at said second end using a second hinge.

2. The knee joint rehabilitation system of claim **1** said goniometer comprising markings disposed on a rod slidably engaged with a sleeve, the goniometer configured so that said user can identify a position of the markings with respect to the sleeve and manually record said position of the lower leg support.

3. The knee joint rehabilitation system of claim **1** said goniometer comprising an electromechanical goniometer configured for providing an electrical signal corresponding to said position of said lower leg support.

4. The knee joint rehabilitation system of claim **1** the lower leg support comprising a top surface shaped to receive a posterior portion of a leg of a user, the lower leg support being contoured to receive a calf muscle of the user.

5. The knee joint rehabilitation system of claim **4**, the lower leg support comprising a first groove and a second groove, said lower leg support increases in thickness along a portion of its length.

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6. The knee joint rehabilitation system of claim 5, said lower leg support comprising a first group of one or more sensors and a second group of one or more sensors;

said first group of one or more sensors being disposed on said first groove; and

said second group of one or more sensors being disposed on said second groove.

7. The knee joint rehabilitation system of claim 6, said lower leg support being configured so that the posterior portion of the leg of said user contacts said first group of one or more sensors at a first angular position of the lower leg support and is urged towards said second group of one or more sensors at a second angular position of the lower leg support.

8. The knee joint rehabilitation system of claim 1 further comprising:

a controller configured for controlling extension and retraction of said actuator when adjusting the position of said lower leg support.

9. The knee joint rehabilitation system of claim 8, said controller configured for handheld control and including a switch or button for controlling said angular position between said lower leg support and said height-adjustment bar.

10. The knee joint rehabilitation system of claim 8 wherein said controller is configured for executing one or more preselected routines for controlling said angular position between said lower leg support and said height-adjustment bar.

11. A knee joint rehabilitation system comprising:

a knee joint flexion device, comprising:

a tubular strut having a first support foot and a bracket attached thereto or configured for attachment to the tubular strut, the tubular strut forming a first aperture;

a height-adjustment bar configured to be translatable disposed in the tubular strut when coupled thereto, the height-adjustment bar forming a plurality of second apertures extending through a diameter of the height-adjustment bar;

a support bar fixedly connected substantially perpendicularly to the height-adjustment bar, the support bar being configured for receiving at least one of a plurality of pads so as to provide a padded support bar, the padded support bar being configured to receive a knee joint posterior;

said plurality of pads being configured to removably mount to the support bar, each pad having an outer diameter different from that of another of said plurality of pads;

a support strut pivotably connected to the bracket when the knee joint flexion device is in an assembled form, the support strut having a second support foot attached thereto, the support strut pivotable away from parallel the tubular strut to form an angle thereto;

a pin configured for removable positioning in one of the plurality of second apertures to fix the height-adjustment bar against translation;

a lower leg support connected to the height-adjustment bar using a first joint so as to allow for changes in an angular position between said lower leg support and said height-adjustment bar when the position of the lower leg support is adjusted; and

a first actuator including a first end and a second end, said actuator being pivotably connected to the lower

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leg support at said first end and pivotably connected to the height-adjustment bar at said second end.

12. The system of claim 11 further comprising a goniometer configured for indicating a position of said lower leg support.

13. The system of claim 11 wherein the lower leg support comprises a surface shaped to receive a posterior portion of a leg of a user and to accommodate for changes in positioning of the leg of the user for different angles of the lower leg support.

14. The system of claim 11, the lower leg support comprising a support pack or panel comprising a first groove configured for receiving a posterior portion of a patient's lower leg,

said lower leg support increasing in thickness along a portion of its length extending from a first end to a second end, the lower leg support comprising a second groove.

15. The system of claim 14 further comprising:

a first group of one or more sensors disposed on the lower leg support within said first groove; and

a second group of one or more sensors disposed on the lower leg support within said second groove.

16. The system of claim 11, further comprising a controller configured for handheld control by a user, the controller including a switch or button for controlling said angular position between said lower leg support and said height-adjustment bar.

17. The system of claim 11, further comprising a controller being configured for executing one or more preselected routines for controlling said angular position between said lower leg support and said height-adjustment bar.

18. A knee joint rehabilitation system comprising:

a height-adjustment bar;

a plurality of support bars, each support bar among the plurality of support bar being configured for removable coupling to the height-adjustment bar, each of the support bars being configured to receive a knee joint posterior;

the plurality of support bars being configured for providing different effective diameters when individual support bars among the plurality of support bars are mounted to the height-adjustment bar; and

a lower leg support connected to the height-adjustment bar using a first joint or configured for connection thereto using said first joint so as to allow for changes in an angular position between said lower leg support and said height-adjustment bar when the position of the lower leg support is adjusted;

an actuator configured for extension and retraction so as to adjust the position of said lower leg support, the actuator including a first end and a second end, the actuator being pivotably connected to the lower leg support at said first end and pivotably connected to the height-adjustment bar at said second end.

19. The knee joint rehabilitation system of claim 18, said lower leg support comprising a first group of one or more sensors and a second group of one or more sensors;

said first group of one or more sensors being disposed on said first groove; and

said second group of one or more sensors being disposed on said second groove.

20. The knee joint rehabilitation system of claim 19, said lower leg support being configured so that the posterior portion of said patient's leg contacts said first group of one or more sensors at a first angular position of the lower leg

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support and is urged towards said second group of one or more sensors at a second angular position of the lower leg support.

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